# Holographic RIS-Aided Reference Energy Detection for Low-Complexity Noncoherent Communications

Miyu Feng, Shiwei Guo, Tiebin Mi, and Robert Caiming Qiu

***Abstract*—This paper presents a comprehensive framework for signal recovery in holographic reconfigurable intelligent surface (RIS) receivers, where only amplitude information is available at the receiver. The proposed system employs a known reference wave to enable phase retrieval from intensity-only measurements. We formulate the complete signal model from transmission to reception, and propose a Gerchberg-Saxton algorithm to estimate the wave vector and complex channel gain. Building upon this, we develop a Gauss-Newton algorithm for efficient signal recovery to conduct real-time estimation of the symbols utilizing the property of rapid convergence. Simulation results validate the effectiveness of the proposed approach, showing accurate signal recovery under various signal-to-noise ratio conditions. The complete modeling framework and algorithm implementation details are provided to facilitate reproducibility and further research in RIS-based communication systems.**

***Index Terms*—Reconfigurable Intelligent Surface, holographic receiver, phase retrieval, signal recovery, amplitude-only measurements.**

## I. Introduction

Reconfigurable intelligent surfaces (RISs) have emerged as a promising technology for next-generation wireless communication systems, owing to their ability to reshape the wireless propagation environment in a programmable and energy-efficient manner [1], [2]. By adaptively controlling the electromagnetic response of a large number of low-cost reflecting elements, RISs can provide additional degrees of freedom for coverage enhancement, interference management, and spatial signal manipulation. More recently, the functionality of RISs has been extended beyond passive beamforming toward active reception and signal processing, giving rise to the concept of holographic RIS receivers [3]. By exploiting densely deployed metasurface elements, holographic RIS receivers are expected to capture rich spatial information of impinging electromagnetic waves while maintaining much lower hardware complexity than conventional antenna arrays equipped with fully digital radio-frequency chains.

Despite these advantages, a fundamental bottleneck of holographic RIS receivers lies in the acquisition of complex-valued signal information. Conventional coherent receivers rely on down-conversion, local oscillator synchronization, and in-phase/quadrature sampling to obtain both amplitude and phase information. In contrast, practical RIS-based or metasurface-based receivers may only have access to intensity or power measurements due to hardware, power, and integration constraints [4]. As a result, the received signal phase is not directly observable, and the recovery of complex-valued symbols from intensity-only measurements naturally leads to a phase retrieval problem, which has been extensively studied in optics and signal processing [5], [6].

A promising way to alleviate this ambiguity is to introduce a known reference wave before intensity detection. Inspired by optical holography, where a reference beam interferes with an object wave to encode both amplitude and phase information into an intensity pattern [7], a reference-assisted RIS receiver superimposes a known radio-frequency reference signal with the unknown received signal at the metasurface or receiver front end. The resulting square-law measurement contains not only the power of the unknown signal but also a reference-induced interference term. This interference term serves as a phase anchor and makes it possible to infer the hidden complex-valued signal from real-valued intensity observations.

[8] investigated the application of phase retrieval techniques for channel estimation in orthogonal frequency-division multiplexing (OFDM) systems, demonstrating that accurate channel estimation is possible even when only magnitude information is available, which shows the phase retrieval problems often appear in the communication area. [9] proposed an affine phase retrieval framework for wireless communication systems, where the received signal is modeled as an affine transformation of the transmitted signal, and developed algorithms for signal recovery under this model, where the problems formulation is very similar to the reference-assisted phase retrieval model in this paper.

However, existing studies on RIS-assisted reception and phase retrieval still leave several key questions insufficiently addressed. Most RIS receiver designs focus on enhancing the received signal power, improving beamforming gain, or reducing hardware complexity, while the intrinsic identifiability of complex symbols under intensity-only measurements has not been fully characterized. In particular, when the receiver observes only the squared magnitude of the superimposed field, different transmitted symbols may produce identical or nearly identical intensity patterns. In such cases, no detector, including maximum-likelihood detection or iterative phase retrieval algorithms, can reliably distinguish the transmitted symbols. Therefore, the central issue is not only how to design an efficient recovery algorithm, but also whether the transmitted symbols are uniquely encoded in the intensity domain in the first place.

This issue becomes more critical in holographic RIS receivers because the measured intensity pattern is jointly deter-

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mined by multiple coupled components, including the reference wave, the pulse-shaping or sampling operation, and the RIS array manifold. If the effective matrix maps two different symbol vectors to the same field response, the corresponding intensity measurements remain indistinguishable regardless of how the detector is implemented. Likewise, even when the effective field responses are different, an improperly designed reference wave may fail to project their phase difference into the observable intensity domain.

Consequently, the lack of a systematic identifiability analysis limits the theoretical understanding and practical design of reference-assisted holographic RIS receivers. Existing phase retrieval methods often assume generic random measurements or focus on continuous signal recovery, whereas communication symbols belong to finite constellations with specific algebraic structures. This discrete nature creates both opportunities and challenges: on one hand, exact recovery may be possible under weaker conditions than those required for general continuous phase retrieval; on the other hand, constellation-dependent ambiguities may still arise if the reference wave and the RIS measurement structure are not properly designed. Therefore, it is essential to investigate how the reference wave, the shaping matrix, and the RIS array manifold jointly determine the uniqueness and stability of symbol recovery.

In this work, we address this problem by developing a reference-assisted phase retrieval framework for holographic RIS receivers. Instead of treating the reference signal merely as an auxiliary hardware component, we characterize its role as a phase-anchoring mechanism that converts hidden phase information into measurable intensity variations. We analyze the discrete identifiability condition of transmitted symbols under the reference-assisted intensity observation model and show that successful recovery requires two complementary properties: the effective RIS measurement matrix must preserve the differences between constellation points, and the reference wave must separate these differences in the intensity domain. Based on this insight, we further introduce an energy-domain minimum distance metric to quantify the distinguishability of different transmitted symbols after square-law detection. This metric provides a direct link between reference-wave design, RIS measurement diversity, and symbol detection performance.

The main contributions of this paper are summarized as follows:

- We establish a reference-assisted holographic RIS receiver model, in which complex-valued communication symbols are recovered from intensity-only measurements through the interference between the unknown signal and a known reference wave.
- We derive discrete identifiability conditions for reference-assisted phase retrieval over finite constellations and reveal the joint impact of the reference wave, the pulse-shaping matrix, and the RIS array manifold on symbol uniqueness.
- We introduce an energy-domain minimum distance metric to characterize the stability of symbol recovery and to guide reference-wave and RIS measurement design.
- We develop and compare maximum-likelihood and low-complexity phase retrieval based detectors, demonstrating that proper reference-wave diversity significantly improves symbol distinguishability and detection performance.

## II. System Model

We consider a holographic RIS receiver system in which $L$ single-antenna users transmit complex baseband symbols to an RIS composed of multiple receiving elements. The RIS senses only the intensity of the interference pattern formed by the received signal and a known reference wave at each of its elements. The known reference signal is generated by a local oscillator [10]. The goal is to estimate the complex channel parameters and degree of arrival (DOA) angles, and subsequently recover the transmitted symbols utilizing these channel information.

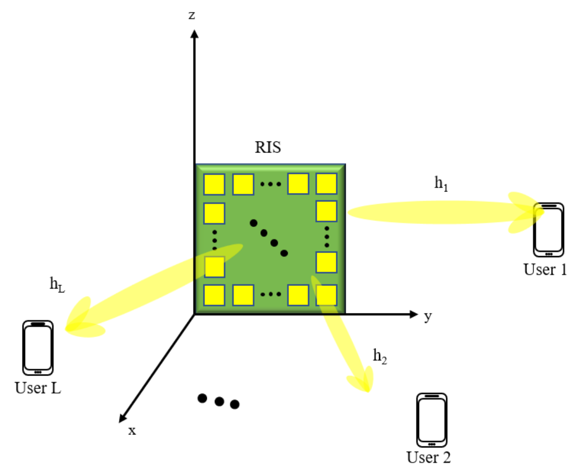


Fig. 1: System model of holographic RIS receiver with amplitude-only measurements.

### A. Transmitter Model

Consider an uplink transmission system with $L$ single-antenna users. The information-bearing symbols transmitted by the $l$-th user within one signaling block are collected as $\mathbf{s}_l = [s_{l,0}, s_{l,1}, \ldots, s_{l,K-1}]^T \in \mathcal{S}^K$, where $K$ denotes the number of symbols in the considered block, $s_{l,k}$ represents the $k$-th transmitted symbol of user $l$, and $\mathcal{S} = c_0, c_1, \ldots, c_{M_{\text{mod}}-1} \subset \mathbb{C}$ denotes the modulation constellation. And the constellation order is given by $|\mathcal{S}| = M_{\text{mod}}$. The constellation points are assumed to be equiprobable and normalized such that $\frac{1}{M_{\text{mod}}} \sum_{q=0}^{M_{\text{mod}}-1} |c_q|^2 = \mathbb{E}|s_{l,k}|^2 = 1$.

After pulse shaping, the complex baseband transmit waveform of the $l$-th user is given by

$$x_l(t) = \sqrt{P_l} \left( \sum_{k=0}^{K-1} s_{l,k} p(t - kT_s) + \xi_l q(t) \right), \tag{1}$$

where $P_l$ is the average transmit power of user $l$, $p(t)$ is the pulse-shaping filter, $T_s$ is the symbol duration, $\xi_l \in \mathbb{C}$ is a deterministic pilot or bias coefficient, and $q(t)$ is a known pilot waveform. The deterministic term $\xi_l q(t)$ is optional and is introduced to allow a nonzero-mean transmitted waveform when required by the subsequent reference-assisted recovery process. If no transmitted pilot or bias is used, we set $\xi_l = 0$.

The corresponding real-valued passband signal is expressed as

$$x_l^{\mathrm{RF}}(t) = \sqrt{2}\Re\left\{x_l(t)e^{j(2\pi f_c t+\varphi_l)}\right\}, \quad l = 1, 2, \ldots, L, \quad (2)$$

where $f_c$ is the carrier frequency and $\varphi_l$ denotes the initial carrier phase offset of the $l$-th user. Let

$$x_l(t) = x_{l,I}(t) + jx_{l,Q}(t), \quad (3)$$

where $x_{l,I}(t)$ and $x_{l,Q}(t)$ are the in-phase and quadrature components of the complex baseband waveform, respectively. Then (2) can be equivalently written as

$$\begin{aligned} x_l^{\mathrm{RF}}(t) = \sqrt{2}\big[&x_{l,I}(t)\cos(2\pi f_c t + \varphi_l) \\ &- x_{l,Q}(t)\sin(2\pi f_c t + \varphi_l)\big]. \end{aligned} \quad (4)$$

For compact notation, the multi-user symbol vector at symbol time $k$ is defined as $\mathbf{s}_k = [s_{1,k}, s_{2,k}, \ldots, s_{L,k}]^T \in \mathcal{S}^L$, and the multi-user complex baseband waveform vector as $\mathbf{x}(t) = [x_1(t), x_2(t), \ldots, x_L(t)]^T \in \mathbb{C}^L$. Using (1), the multi-user baseband waveform can be written as

$$\mathbf{x}(t) = \mathbf{D}_P\left(\sum_{k=0}^{K-1}\mathbf{s}_k p(t-kT_s) + \boldsymbol{\xi} q(t)\right), \quad (5)$$

where $\mathbf{D}_P = \mathrm{diag}(\sqrt{P_1}, \sqrt{P_2}, \ldots, \sqrt{P_L})$ is the transmit-power scaling matrix, and $\boldsymbol{\xi} = [\xi_1, \xi_2, \ldots, \xi_L]^T \in \mathbb{C}^L$ collects the deterministic pilot or bias coefficients of all users.

To facilitate the subsequent symbol-level analysis, we distinguish three time scales. Each symbol has duration $T_s$ and is observed through $M$ consecutive detection subslots, each of duration

$$\Delta_t = \frac{T_s}{M}. \quad (6)$$

The $M$ subslots associated with one symbol constitute a *symbol observation block*, whereas the signaling block contains $K$ such symbol observation blocks. The center of the $m$-th subslot of symbol $k$ is

$$t_{k,m} = kT_s + \left(m + \frac{1}{2}\right)\Delta_t, \quad {\scriptstyle k=0,1,\ldots,K-1, \atop m=0,1,\ldots,M-1}. \quad (7)$$

Thus, the signaling block contains $N_t = KM$ baseband observation instants. For notational compatibility with a single time index, we use $n = kM + m$ and $t_n = t_{k,m}$. For the $l$-th user, define the sampled baseband transmit vector as

$$\tilde{\mathbf{x}}_l = [x_l(t_0), x_l(t_1), \ldots, x_l(t_{N_t-1})]^T \in \mathbb{C}^{N_t}. \quad (8)$$

From (1), we obtain

$$\tilde{\mathbf{x}}_l = \sqrt{P_l}\left(\mathbf{P}\mathbf{s}_l + \xi_l\mathbf{q}\right), \quad (9)$$

where $\mathbf{P} \in \mathbb{C}^{KM\times K}$ denotes the pulse-shaping matrix with entries $[\mathbf{P}]_{kM+m,j} = p(t_{k,m} - jT_s)$, and $\mathbf{q} \in \mathbb{C}^{KM}$ denotes the pilot waveform sampled at the same subslot centers. This two-index representation makes explicit that several waveform samples are associated with each transmitted symbol; it does not identify an individual sample with a complete symbol observation.

Stacking the sampled baseband transmit vectors of all users gives

$$\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_1^T, \tilde{\mathbf{x}}_2^T, \ldots, \tilde{\mathbf{x}}_L^T]^T \in \mathbb{C}^{LKM}. \quad (10)$$

Similarly, define the stacked transmitted symbol vector as

$$\mathbf{s} = [\mathbf{s}_1^T, \mathbf{s}_2^T, \ldots, \mathbf{s}_L^T]^T \in \mathcal{S}^{LK}. \quad (11)$$

Then the sampled multi-user transmit waveform can be represented as

$$\begin{aligned} \tilde{\mathbf{x}} &= (\mathbf{D}_P \otimes \mathbf{P})\mathbf{s} + (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q}, \\ &= \mathbf{P}_{\mathrm{tx}}\mathbf{s} + \mathbf{x}_0. \end{aligned} \quad (12)$$

where $\mathbf{P}_{\mathrm{tx}} = \mathbf{D}_P \otimes \mathbf{P} \in \mathbb{C}^{LKM\times LK}$, $\mathbf{x}_0 = (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q} \in \mathbb{C}^{LKM}$, and $\otimes$ denotes the Kronecker product. Thus, the transmitter-side equivalent discrete model is compactly written as

$$\tilde{\mathbf{x}} = \mathbf{P}_{\mathrm{tx}}\mathbf{s} + \mathbf{x}_0. \quad (13)$$

The matrix $\mathbf{P}_{\mathrm{tx}}$ characterizes the linear mapping from the transmitted constellation symbols to the $M$ waveform samples within each symbol observation block. This representation allows the subsequent receiver model to incorporate pulse shaping, spatial propagation, and holographic RIS observations through an equivalent measurement matrix while keeping the analysis in the discrete symbol domain.

### B. Far-Field Channel Model

As shown in Fig. 1, we consider a holographic RIS receiver composed of a uniform planar array placed on the $yz$-plane. The array consists of $N_y$ elements along the $y$-axis and $N_z$ elements along the $z$-axis, resulting in a total of

$$N_{\mathrm{r}} = N_y N_z \quad (14)$$

receiving elements. The position vector of the $(n, m)$-th element in the RIS array is defined as

$$\mathbf{r}_{n,m} = [0,\, y_n,\, z_m]^{\mathrm{T}}, \quad (15)$$

where $n = 1, \ldots, N_y$ and $m = 1, \ldots, N_z$, and $y_n = \left(n - \frac{N_y+1}{2}\right)d_y$, $\quad z_m = \left(m - \frac{N_z+1}{2}\right)d_z$. Here, $d_y$ and $d_z$ denote the inter-element spacings along the $y$- and $z$-axes, respectively. The centered coordinate system is adopted for notational convenience; using a corner element as the origin leads only to a user-dependent common phase shift, which can be absorbed into the complex channel gain.

Under the far-field assumption, the impinging signal from each user can be modeled as a plane wave across the RIS aperture. Let $\theta_l$ and $\phi_l$ denote the angle of arrival (AoA) of the $l$-th user, where $\theta_l$ is the polar angle with respect to the array broadside, i.e., the positive $x$-axis, and $\phi_l$ is the azimuth angle in the $yz$-plane measured from the positive $y$-axis. The corresponding wave vector is given by

$$\mathbf{k}_l = [k_{l,x}, k_{l,y}, k_{l,z}]^T, \quad (16)$$

with $k_{l,x} = k_0\cos\theta_l$, $\quad k_{l,y} = k_0\sin\theta_l\cos\phi_l$, $\quad k_{l,z} = k_0\sin\theta_l\sin\phi_l$, $k_0 = \frac{2\pi}{\lambda}$ is the free-space wavenumber and $\lambda$ is the carrier wavelength.

Following the plane-wave model, the phase of the incident field from user $l$ at the $(n,m)$-th RIS element is determined by the projection of the wave vector onto the element position, i.e.,

$$\psi_{n,m}^{(l)} = \mathbf{k}_l^T \mathbf{r}_{n,m} = k_{l,y} y_n + k_{l,z} z_m. \tag{17}$$

Since the RIS aperture is assumed to be sufficiently small compared with the propagation distance, the amplitude variation of the incident wave across different RIS elements is neglected for each user. Therefore, the spatial variation across the RIS elements is mainly characterized by the phase profile in (17), while the large-scale path loss, shadowing, and user-dependent phase offset are incorporated into a complex channel gain.

Accordingly, the channel vector from the $l$-th user to all RIS receiving elements is modeled as

$$\mathbf{h}_l = \alpha_l \mathbf{v}_l \in \mathbb{C}^{N_\mathrm{r} \times 1}, \tag{18}$$

where $\alpha_l \in \mathbb{C}$ denotes the complex channel gain of user $l$, and $\mathbf{v}_l$ is the normalized array steering vector. The element of $\mathbf{v}_l$ corresponding to the $(n,m)$-th RIS element is given by

$$[\mathbf{v}_l]_{\iota(n,m)} = e^{j(k_{l,y} y_n + k_{l,z} z_m)}, \tag{19}$$

where $\iota(n,m) = (n-1)N_z + m$ maps the two-dimensional element index $(n,m)$ to the one-dimensional vector index.

Equivalently, the array steering vector can be written as the Kronecker product of two one-dimensional steering vectors:

$$\mathbf{v}_l = \mathbf{a}_y(\theta_l, \phi_l) \otimes \mathbf{a}_z(\theta_l, \phi_l), \tag{20}$$

where $\otimes$ denotes the Kronecker product,

$$\mathbf{a}_y(\theta_l, \phi_l) = \left[e^{jk_{l,y} y_1}, e^{jk_{l,y} y_2}, \ldots, e^{jk_{l,y} y_{N_y}}\right]^T \in \mathbb{C}^{N_y \times 1}, \tag{21}$$

and

$$\mathbf{a}_z(\theta_l, \phi_l) = \left[e^{jk_{l,z} z_1}, e^{jk_{l,z} z_2}, \ldots, e^{jk_{l,z} z_{N_z}}\right]^T \in \mathbb{C}^{N_z \times 1}. \tag{22}$$

Collecting the channel vectors of all $L$ users, the multi-user channel matrix between the users and the RIS receiver is expressed as

$$\begin{aligned} \mathbf{H} &= [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_L] \in \mathbb{C}^{N_\mathrm{r} \times L} \\ &= [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_L] \operatorname{diag}(\alpha_1, \alpha_2, \ldots, \alpha_L), \end{aligned} \tag{23}$$

where the second line follows from the decomposition of each user channel vector in (18). For compactness, define $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_L] \in \mathbb{C}^{N_\mathrm{r} \times L}$ as the RIS array manifold matrix, and $\mathbf{\Gamma} = \operatorname{diag}(\alpha_1, \alpha_2, \ldots, \alpha_L) \in \mathbb{C}^{L \times L}$ as the diagonal matrix of complex channel gains. Then the multi-user channel matrix is compactly written as

$$\mathbf{H} = \mathbf{V}\mathbf{\Gamma}. \tag{24}$$

This decomposition separates the directional spatial response of the RIS, represented by $\mathbf{V}$, from the user-dependent complex propagation gains, represented by $\mathbf{\Gamma}$. It will be used in the subsequent receiver model to characterize how the pulse-shaped user signals are spatially mapped onto the holographic RIS elements.

### C. Reference-Assisted Holographic Reception Model

At the receiver side, the holographic RIS collects the superimposed signals from all users through its $N_\mathrm{r}$ receiving elements. Using the multi-user channel matrix $\mathbf{H} \in \mathbb{C}^{N_\mathrm{r} \times L}$ defined in the previous subsection, the complex baseband signal impinging on the RIS at time $t$ is written as

$$\mathbf{r}(t) = \mathbf{H}\mathbf{x}(t). \tag{25}$$

To enable phase information recovery from intensity-only observations, a known reference wave is introduced at the RIS receiver. Let $\mathbf{b} = [b_1, b_2, \ldots, b_{N_\mathrm{r}}]^T \in \mathbb{C}^{N_\mathrm{r}}$ denote the equivalent complex baseband reference vector, where $b_i = \rho_i e^{j\varphi_i}, i = 1, 2, \ldots, N_\mathrm{r}$. Here, $\rho_i$ and $\varphi_i$ are the known amplitude and phase of the reference wave injected at the $i$-th RIS element, respectively. The corresponding passband reference signal is

$$b_i^{\mathrm{RF}}(t) = \sqrt{2}\Re\left\{b_i e^{j2\pi f_c t}\right\} = \sqrt{2}\rho_i \cos(2\pi f_c t + \varphi_i). \tag{26}$$

After superimposing the received signal and the reference wave, the equivalent complex baseband field at the RIS is given by

$$\mathbf{u}(t) = \mathbf{r}(t) + \mathbf{b} = \mathbf{H}\mathbf{x}(t) + \mathbf{b}. \tag{27}$$

Equivalently, the real-valued RF signal at the $i$-th RIS element can be expressed as

$$u_i^{\mathrm{RF}}(t) = \sqrt{2}\Re\left\{u_i(t) e^{j2\pi f_c t}\right\} + \nu_i^{\mathrm{RF}}(t), \tag{28}$$

where $u_i(t)$ is the $i$-th element of $\mathbf{u}(t)$ and $\nu_i^{\mathrm{RF}}(t)$ denotes the RF noise.

The holographic RIS receiver is assumed to employ square-law detection at each element. The $m$-th detection subslot of symbol $k$ is the interval

$$\mathcal{I}_{k,m} = [kT_s + m\Delta_t,\, kT_s + (m+1)\Delta_t). \tag{29}$$

Consequently, the detector output at the $i$-th RIS element is the normalized energy (or average intensity) over this subslot:

$$z_i[k,m] = \frac{1}{\Delta_t} \int_{\mathcal{I}_{k,m}} \left|u_i^{\mathrm{RF}}(\tau)\right|^2 d\tau. \tag{30}$$

After carrier-frequency components are removed by low-pass filtering, the corresponding baseband representation is

$$z_i[k,m] = \frac{1}{\Delta_t} \int_{\mathcal{I}_{k,m}} |[\mathbf{H}\mathbf{x}(\tau)]_i + b_i|^2 d\tau + w_i[k,m]. \tag{31}$$

Thus, each detector output is a normalized energy sample obtained by integration over a finite subslot, rather than an instantaneous RF sample. If the complex envelope is approximately constant within $\mathcal{I}_{k,m}$, (31) reduces to

$$z_i[k,m] \simeq |[\mathbf{H}\mathbf{x}(t_{k,m})]_i + b_i|^2 + w_i[k,m]. \tag{32}$$

Stacking all RIS-element outputs in subslot $(k,m)$ gives

$$\mathbf{z}[k,m] = |\mathbf{H}\mathbf{x}[k,m] + \mathbf{b}|^2 + \mathbf{w}[k,m] \in \mathbb{R}^{N_\mathrm{r}}, \tag{33}$$

where $\mathbf{x}[k,m] \triangleq \mathbf{x}(t_{k,m})$, and the squared modulus is applied element-wise. The complete observation associated with symbol $k$ is then

$$\mathbf{z}_k = \operatorname{col}_{m=0}^{M-1}\{\mathbf{z}[k,m]\} \in \mathbb{R}^{MN_\mathrm{r}}. \tag{34}$$

Therefore, symbol recovery uses jointly the $MN_{\mathrm{r}}$ energy samples collected across all RIS elements and all $M$ detection subslots in the symbol observation block. These samples are retained as a vector rather than summed into a single scalar, because their spatial and temporal variations provide the information required for symbol identifiability.

For the general pulse-shaped waveform,

$$\mathbf{x}[k,m] = \mathbf{D}_P \left( \sum_{j=0}^{K-1} \mathbf{s}_j p(t_{k,m} - jT_s) + \boldsymbol{\xi} q(t_{k,m}) \right), \quad (35)$$

so $\mathbf{z}_k$ can depend on neighboring symbols when the pulse has intersymbol overlap. In this general case, all $KMN_{\mathrm{r}}$ energy samples in the signaling block are stacked and the symbols are detected jointly. Let $\boldsymbol{\Pi}$ denote the permutation matrix that converts the user-major stacking in $\tilde{\mathbf{x}}$ to time-major stacking. Using (13), the complete signaling-block observation is

$$\mathbf{z} = |\mathbf{G}\mathbf{s} + \mathbf{b}_{\mathrm{eq}}|^2 + \mathbf{w}, \quad (36)$$

where

$$\mathbf{G} = (\mathbf{I}_{KM} \otimes \mathbf{H})\boldsymbol{\Pi}\mathbf{P}_{\mathrm{tx}}, \quad (37)$$

$$\mathbf{b}_{\mathrm{eq}} = \mathbf{1}_{KM} \otimes \mathbf{b} + (\mathbf{I}_{KM} \otimes \mathbf{H})\boldsymbol{\Pi}\mathbf{x}_0. \quad (38)$$

Here, $\mathbf{s} \in \mathcal{S}^{LK}$, $\mathbf{z}, \mathbf{w} \in \mathbb{R}^{KMN_{\mathrm{r}}}$, and $\mathbf{G} \in \mathbb{C}^{KMN_{\mathrm{r}} \times LK}$. The compact model in (36) will serve as the basis for the subsequent identifiability analysis and detector design.

If intersymbol overlap is negligible, the $k$-th symbol observation vector depends only on $\mathbf{s}_k$ and admits the symbol-wise model

$$\mathbf{z}_k = |\mathbf{G}_k \mathbf{s}_k + \mathbf{b}_k|^2 + \mathbf{w}_k, \quad (39)$$

where $\mathbf{G}_k \in \mathbb{C}^{MN_{\mathrm{r}} \times L}$ and $\mathbf{b}_k \in \mathbb{C}^{MN_{\mathrm{r}}}$ collect the spatial responses and known offsets over the $M$ subslots. The corresponding finite-alphabet detector is

$$\widehat{\mathbf{s}}_k = \arg\min_{\mathbf{a} \in \mathcal{S}^L} \left\| \mathbf{z}_k - |\mathbf{G}_k \mathbf{a} + \mathbf{b}_k|^2 \right\|_2^2. \quad (40)$$

Hence, symbol-wise recovery is valid under negligible intersymbol overlap; otherwise, the block-level model (36) must be used for joint sequence detection.

### *D. Two-Phase Transmission Protocol and Unified Observation Model*

Based on the reference-assisted holographic reception model, we consider a two-phase transmission protocol consisting of a training phase and a data transmission phase. In the training phase, the users transmit known pilot waveforms, and the RIS receiver estimates the channel from intensity-only measurements. In the data transmission phase, the users transmit information-bearing symbols, and the receiver detects the transmitted symbols based on the estimated channel. Both phases follow the same reference-assisted intensity observation principle, but the unknown variables are different.

*1) Training Phase:* During the training phase, all users transmit known pilot signals. Let $\mathbf{x}_{\mathrm{p}}[t] \in \mathbb{C}^L$ denote the known multi-user waveform sample at the $t$-th pilot detection subslot, where $t = 0, 1, \ldots, T_{\mathrm{p}} - 1$ and $T_{\mathrm{p}}$ is the total number of pilot energy observations per RIS element. For example, if the training block contains $K_{\mathrm{p}}$ pilot symbols and each symbol uses $M$ detection subslots, then $T_{\mathrm{p}} = K_{\mathrm{p}} M$. The corresponding reference vector is denoted by $\mathbf{b}_{\mathrm{p}}[t] \in \mathbb{C}^{N_{\mathrm{r}}}$. According to (33), the received intensity measurement during the training phase for detection-subslot indices $t = 0, 1, \ldots, T_{\mathrm{p}} - 1$ is given by

$$\mathbf{z}_{\mathrm{p}}[t] = |\mathbf{H}\mathbf{x}_{\mathrm{p}}[t] + \mathbf{b}_{\mathrm{p}}[t]|^2 + \mathbf{w}_{\mathrm{p}}[t]. \quad (41)$$

where $\mathbf{z}_{\mathrm{p}}[t] \in \mathbb{R}^{N_{\mathrm{r}}}$ is the normalized-energy observation vector, and $\mathbf{w}_{\mathrm{p}}[t] \in \mathbb{R}^{N_{\mathrm{r}}}$ denotes the effective post-detection noise.

For compact representation, define the vectorized channel as

$$\mathbf{h} = \mathrm{vec}(\mathbf{H}) \in \mathbb{C}^{N_{\mathrm{r}} L}. \quad (42)$$

Using the identity

$$\mathbf{H}\mathbf{x}_{\mathrm{p}}[t] = \left( \mathbf{x}_{\mathrm{p}}^T[t] \otimes \mathbf{I}_{N_{\mathrm{r}}} \right) \mathbf{h}, \quad (43)$$

(41) can be rewritten as

$$\mathbf{z}_{\mathrm{p}}[t] = |\mathbf{G}_{\mathrm{p}}[t]\mathbf{h} + \mathbf{b}_{\mathrm{p}}[t]|^2 + \mathbf{w}_{\mathrm{p}}[t], \quad (44)$$

where $\mathbf{G}_{\mathrm{p}}[t] = \mathbf{x}_{\mathrm{p}}^T[t] \otimes \mathbf{I}_{N_{\mathrm{r}}} \in \mathbb{C}^{N_{\mathrm{r}} \times N_{\mathrm{r}} L}$. Stacking all pilot observations yields the block-level training model

$$\mathbf{z}_{\mathrm{p}} = |\mathbf{G}_{\mathrm{p}}\mathbf{h} + \mathbf{b}_{\mathrm{p}}|^2 + \mathbf{w}_{\mathrm{p}}, \quad (45)$$

where $\mathbf{z}_{\mathrm{p}}$, $\mathbf{b}_{\mathrm{p}}$, and $\mathbf{w}_{\mathrm{p}}$ collect the corresponding quantities over all pilot observations, and $\mathbf{G}_{\mathrm{p}} = \left[\mathbf{G}_{\mathrm{p}}[0], \mathbf{G}_{\mathrm{p}}[1], \ldots, \mathbf{G}_{\mathrm{p}}[T_{\mathrm{p}} - 1]\right]^{\mathrm{T}}$. If the far-field channel structure in (24) is exploited, the unknown channel vector $\mathbf{h}$ can be further parameterized by a lower-dimensional channel parameter vector, such as the complex path-gain vector when the angles of arrival are known or pre-estimated.

*2) Data Transmission Phase:* After the training phase, the receiver obtains an estimate of the channel, denoted by $\hat{\mathbf{H}}$. In the data transmission phase, the users transmit unknown information-bearing symbols. Let $\mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{LK_{\mathrm{d}}}$ denote the stacked data-symbol vector within one data block, where $K_{\mathrm{d}}$ is the number of data symbols per user. Following the transmitter-side equivalent model, the sampled data waveform can be expressed as

$$\tilde{\mathbf{x}}_{\mathrm{d}} = \mathbf{P}_{\mathrm{tx},d}\mathbf{s}_{\mathrm{d}} + \mathbf{x}_{0,d}, \quad (46)$$

where $\mathbf{P}_{\mathrm{tx},d} \in \mathbb{C}^{LMK_{\mathrm{d}} \times LK_{\mathrm{d}}}$ is the data-phase pulse-shaping matrix and $\mathbf{x}_{0,d} \in \mathbb{C}^{LMK_{\mathrm{d}}}$ is the deterministic transmit-side pilot or bias component, if present.

Using the estimated channel $\hat{\mathbf{H}}$, the block-level data-phase intensity observation can be written as

$$\mathbf{z}_{\mathrm{d}} = \left| \mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})\mathbf{s}_{\mathrm{d}} + \mathbf{b}_{\mathrm{d}} \right|^2 + \mathbf{w}_{\mathrm{d}}, \quad (47)$$

where $\mathbf{z}_{\mathrm{d}}, \mathbf{w}_{\mathrm{d}} \in \mathbb{R}^{Q_{\mathrm{d}}}$, $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}}) \in \mathbb{C}^{Q_{\mathrm{d}} \times LK_{\mathrm{d}}}$, and $\mathbf{b}_{\mathrm{d}} \in \mathbb{C}^{Q_{\mathrm{d}}}$, with $Q_{\mathrm{d}} = N_{\mathrm{r}} M K_{\mathrm{d}}$ denoting the total number of scalar intensity observations collected in one data block. The equivalent data-phase measurement matrix $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$ is determined by the estimated channel, the pulse-shaping matrix, and the RIS observation structure. The vector $\mathbf{b}_{\mathrm{d,eq}}$ denotes the equivalent known offset in the data phase, which includes the receiver-side reference wave and the contribution of any deterministic transmit-side bias. The vector $\mathbf{w}_{\mathrm{d}}$ represents the effective post-detection noise.

Equations (45) and (47) show that both channel estimation and data detection can be written in the common reference-assisted intensity measurement form

$$\mathbf{z} = \left|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}_{\mathrm{eq}}\right|^2 + \mathbf{w}. \tag{48}$$

In the training phase, the unknown variable is the channel-related parameter, e.g., $\boldsymbol{\theta} = \mathbf{h}$ or a lower-dimensional parameterization of $\mathbf{H}$. In the data transmission phase, the unknown variable is the finite-alphabet symbol vector, i.e., $\boldsymbol{\theta} = \mathbf{s}_{\mathrm{d}}$. Therefore, the two phases share the same phaseless observation structure, while their identifiability conditions and recovery algorithms differ due to the different structures of the unknown variables. The unified model in (48) will serve as the basis for the subsequent identifiability analysis and detector design.

## III. Uniqueness and Stability of Reference-Assisted Energy Detection

This section studies when the reference-assisted square-law receiver has a unique inverse and when that inverse is robust to perturbations. Following the distinction between injectivity and stability in phase retrieval [11], we do not use matrix rank alone as a surrogate for recoverability. Instead, we derive an exact pairwise condition and lower and upper Lipschitz bounds for the nonlinear energy mapping. The analysis applies to both continuous channel estimation and finite-alphabet symbol detection.

### A. Unified Affine Phase-Retrieval Model

The training- and data-phase observations derived in Section II share the form

$$\mathbf{z} = \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) + \mathbf{w}, \qquad \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) \triangleq \left|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}\right|^2, \tag{49}$$

where $Q$ is the total number of scalar intensity observations, $D$ is the complex dimension of the unknown vector, $\mathbf{G} \in \mathbb{C}^{Q\times D}$, $\boldsymbol{\theta} \in \mathcal{X} \subseteq \mathbb{C}^D$, $\mathbf{b} \in \mathbb{C}^Q$, and $\mathbf{z}, \mathbf{w} \in \mathbb{R}^Q$. Here, $\mathcal{X}$ denotes the feasible set of the unknown parameter, and the modulus and square are applied element-wise. This is an affine phase-retrieval model [9]. For full-channel estimation in the training phase, $\boldsymbol{\theta} = \mathbf{h} = \mathrm{vec}(\mathbf{H})$, $Q = N_{\mathrm{r}}T_{\mathrm{p}}$, and $D = N_{\mathrm{r}}L$. If the channel is represented by a lower-dimensional linear complex parameter, $D$ is replaced by the dimension of that parameter. In the data phase, $\boldsymbol{\theta} = \mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{LK_{\mathrm{d}}}$, $Q = Q_{\mathrm{d}}$, and $D = LK_{\mathrm{d}}$.

The reference term changes the ambiguity class fundamentally. If $\mathbf{b} = 0$, then

$$\mathcal{T}_{\mathbf{G},\mathbf{0}}(e^{j\varphi}\boldsymbol{\theta}) = \mathcal{T}_{\mathbf{G},\mathbf{0}}(\boldsymbol{\theta}), \tag{50}$$

and uniqueness can only be defined modulo a global phase. If $\mathbf{b} \neq 0$, the expansion

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) = |\mathbf{G}\boldsymbol{\theta}|^2 + |\mathbf{b}|^2 + 2\,\mathrm{Re}\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\} \tag{51}$$

contains phase-sensitive interference, so absolute uniqueness on $\mathcal{X}$ becomes possible. A nonzero reference is not sufficient by itself, however: its phases must probe every feasible field difference in nondegenerate real directions.

To quantify this statement, define the realification

$$\boldsymbol{\theta}_{\mathbb{R}} \triangleq \left[\mathrm{Re}\{\boldsymbol{\theta}\}^T \quad \mathrm{Im}\{\boldsymbol{\theta}\}^T\right]^T \in \mathbb{R}^{2D} \tag{52}$$

and, for $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$, the energy-domain distance

$$d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \triangleq \left\|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2)\right\|_2. \tag{53}$$

The optimal lower and upper stability bounds on $\mathcal{X}$ are

$$\alpha_{\mathcal{X}} \triangleq \inf_{\substack{\boldsymbol{\theta}_1,\boldsymbol{\theta}_2\in\mathcal{X}\\ \boldsymbol{\theta}_1\neq\boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\|_2}, \tag{54}$$

$$\beta_{\mathcal{X}} \triangleq \sup_{\substack{\boldsymbol{\theta}_1,\boldsymbol{\theta}_2\in\mathcal{X}\\ \boldsymbol{\theta}_1\neq\boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\|_2}. \tag{55}$$

Thus, the inverse problem is globally stable on $\mathcal{X}$ if $0 < \alpha_{\mathcal{X}} \leq \beta_{\mathcal{X}} < \infty$. The lower bound is the important one: it excludes pairs that are far apart in parameter space but arbitrarily close in energy space.

### B. Exact Secant Identity and Global Uniqueness

The quadratic structure of (49) gives an exact identity, which is stronger than a first-order approximation. For a point $\boldsymbol{\vartheta} \in \mathbb{C}^D$, define the real Jacobian

$$\begin{aligned}\mathbf{J}(\boldsymbol{\vartheta}) \triangleq 2\big[&\,\mathrm{Re}\{\mathrm{diag}[(\mathbf{G}\boldsymbol{\vartheta}+\mathbf{b})^*]\mathbf{G}\},\\ &-\mathrm{Im}\{\mathrm{diag}[(\mathbf{G}\boldsymbol{\vartheta}+\mathbf{b})^*]\mathbf{G}\}\big] \in \mathbb{R}^{Q\times 2D}.\end{aligned} \tag{56}$$

**Lemma 1** (Exact midpoint factorization)**.** *For any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathbb{C}^D$, let $\boldsymbol{\mu} = (\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2$ and $\boldsymbol{\delta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$. Then*

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2) = \mathbf{J}(\boldsymbol{\mu})\boldsymbol{\delta}_{\mathbb{R}}. \tag{57}$$

*Proof.* Applying $|a|^2 - |c|^2 = \mathrm{Re}\{(a+c)^*(a-c)\}$ element-wise gives

$$\mathrm{Re}\big\{[\mathbf{G}(\boldsymbol{\theta}_1+\boldsymbol{\theta}_2)+2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1-\boldsymbol{\theta}_2)\big\}, \tag{58}$$

which equals the right-hand side of (57) after realification. □

**Theorem 1** (Pairwise uniqueness criterion)**.** *The mapping $\mathcal{T}_{\mathbf{G},\mathbf{b}}$ is injective on $\mathcal{X}$ if and only if, for every distinct $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}$,*

$$\mathbf{J}\left(\frac{\boldsymbol{\theta}_1+\boldsymbol{\theta}_2}{2}\right)(\boldsymbol{\theta}_1-\boldsymbol{\theta}_2)_{\mathbb{R}} \neq \mathbf{0}. \tag{59}$$

*A necessary condition, independent of the reference, is*

$$\begin{aligned}&\mathrm{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} = \varnothing,\\ &\mathcal{D}_{\mathcal{X}} \triangleq \{\boldsymbol{\theta}_1-\boldsymbol{\theta}_2 : \boldsymbol{\theta}_1,\boldsymbol{\theta}_2 \in \mathcal{X},\, \boldsymbol{\theta}_1\neq\boldsymbol{\theta}_2\}.\end{aligned} \tag{60}$$

*Proof.* The first claim follows directly from Lemma 1. For the second, if $\mathbf{G}(\boldsymbol{\theta}_1-\boldsymbol{\theta}_2) = 0$, then the two complex fields, and hence their intensities for every $\mathbf{b}$, are identical. □

Theorem 1 makes clear why $\mathrm{rank}(\mathbf{G}) = D$ is necessary for a full-dimensional continuous parameter set but is not sufficient for phase retrieval: even when $\mathbf{G}\boldsymbol{\delta} \neq 0$, its real projection determined jointly by the midpoint field and the reference may vanish. For a finite constellation, full column rank is not necessary; only the finite difference set in (60) must avoid the null space.

Let
$$\mathcal{M}_{\mathcal{X}} \triangleq \{(\boldsymbol{\theta}_1+\boldsymbol{\theta}_2)/2 : \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}\} \tag{61}$$
be the midpoint set. The exact identity immediately yields computable two-sided bounds.

**Proposition 1** (Jacobian stability bounds)**.** *Define*
$$\underline{\sigma}_J \triangleq \inf_{\boldsymbol{\mu}\in\mathcal{M}_{\mathcal{X}}} \sigma_{\min}(\mathbf{J}(\boldsymbol{\mu})), \qquad \overline{\sigma}_J \triangleq \sup_{\boldsymbol{\mu}\in\mathcal{M}_{\mathcal{X}}} \sigma_{\max}(\mathbf{J}(\boldsymbol{\mu})). \tag{62}$$
*Then*
$$\underline{\sigma}_J \,\|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\|_2 \le d_E(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2) \le \overline{\sigma}_J \,\|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\|_2 \tag{63}$$
*for all pairs in $\mathcal{X}$. Consequently, $\alpha_{\mathcal{X}} \ge \underline{\sigma}_J$ and $\beta_{\mathcal{X}} \le \overline{\sigma}_J$. If $\|\boldsymbol{\theta}\|_2 \le R_{\mathcal{X}}$ on $\mathcal{X}$, then*
$$\beta_{\mathcal{X}} \le 2\|\mathbf{G}\|_2 \left(\|\mathbf{G}\|_2 R_{\mathcal{X}} + \|\mathbf{b}\|_\infty\right). \tag{64}$$

*Proof.* The first statement follows by applying the singular-value inequalities to (57). For the explicit upper bound, use $\|\operatorname{diag}(\mathbf{u}^*)\mathbf{G}\|_2 \le \|\mathbf{u}\|_\infty\|\mathbf{G}\|_2$ and $\|\mathbf{G}\boldsymbol{\mu}+\mathbf{b}\|_\infty \le \|\mathbf{G}\|_2 R_{\mathcal{X}} + \|\mathbf{b}\|_\infty$. □

The bound $\underline{\sigma}_J > 0$ is a convenient sufficient condition for global stability, but it can be conservative because it tests every direction at every midpoint, including directions that may not connect two feasible points. The exact constant in (54), or equivalently the restricted secants in (57), is the sharp criterion. Notice also that injectivity alone need not give a useful uniform lower bound on a continuous set; this is precisely the distinction between uniqueness and stability emphasized in [11].

### C. Continuous Channel Identifiability

For a continuous channel parameter, local identifiability at $\boldsymbol{\theta}$ requires
$$\operatorname{rank}(\mathbf{J}(\boldsymbol{\theta})) = 2D, \qquad Q \ge 2D. \tag{65}$$
The smallest singular value determines the local inverse sensitivity:
$$\liminf_{\|\boldsymbol{\delta}\|_2\to 0} \frac{d_E(\boldsymbol{\theta}+\boldsymbol{\delta},\boldsymbol{\theta})}{\|\boldsymbol{\delta}\|_2} = \sigma_{\min}(\mathbf{J}(\boldsymbol{\theta})). \tag{66}$$
Local full rank does not rule out a distant ambiguous channel. Global uniqueness still requires (59) for all pairs, while a positive $\alpha_{\mathcal{X}}$ rules out both exact and near-ambiguities.

These constants also have a direct error interpretation. Let $\boldsymbol{\theta}_\star \in \mathcal{X}$ generate (49), and let $\widehat{\boldsymbol{\theta}}$ be a global minimizer of the intensity least-squares criterion. If $\alpha_{\mathcal{X}} > 0$, then
$$\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_\star\|_2 \le \frac{2\|\mathbf{w}\|_2}{\alpha_{\mathcal{X}}}. \tag{67}$$
Indeed, optimality gives a residual no larger than $\|\mathbf{w}\|_2$; the triangle inequality followed by the lower Lipschitz bound gives (67).

For full-channel training,
$$\mathbf{G}_{\mathrm{p}} = \mathbf{X}_{\mathrm{p}} \otimes \mathbf{I}_{N_{\mathrm{r}}}, \qquad \mathbf{X}_{\mathrm{p}} \in \mathbb{C}^{T_{\mathrm{p}}\times L}, \tag{68}$$
and therefore
$$\begin{aligned} \operatorname{rank}(\mathbf{G}_{\mathrm{p}}) &= N_{\mathrm{r}} \operatorname{rank}(\mathbf{X}_{\mathrm{p}}), \\ \sigma_{\min}(\mathbf{G}_{\mathrm{p}}) &= \sigma_{\min}(\mathbf{X}_{\mathrm{p}}), \\ \kappa(\mathbf{G}_{\mathrm{p}}) &= \kappa(\mathbf{X}_{\mathrm{p}}). \end{aligned} \tag{69}$$
Thus, $\operatorname{rank}(\mathbf{X}_{\mathrm{p}}) = L$ and $T_{\mathrm{p}} \ge L$ are necessary to preserve the complex channel field. They are not sufficient for the intensity mapping. With one real intensity per RIS element and pilot time, $Q = N_{\mathrm{r}}T_{\mathrm{p}}$ and $D = N_{\mathrm{r}}L$, so the dimension condition in (65) strengthens the counting requirement to
$$T_{\mathrm{p}} \ge 2L. \tag{70}$$
Even when this count is met, the pilot and reference phases must make $\mathbf{J}_{\mathrm{p}}$ full column rank and well conditioned.

If the far-field model $\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}$ is used and the angles are known, the unknown can be reduced to $\boldsymbol{\alpha} = [\alpha_1,\ldots,\alpha_L]^T$, with
$$\mathbf{G}_\alpha = \begin{bmatrix} \mathbf{V}\operatorname{diag}(\mathbf{x}_{\mathrm{p}}[0]) \\ \vdots \\ \mathbf{V}\operatorname{diag}(\mathbf{x}_{\mathrm{p}}[T_{\mathrm{p}}-1]) \end{bmatrix}. \tag{71}$$
The necessary count becomes $N_{\mathrm{r}}T_{\mathrm{p}} \ge 2L$. In addition, $\mathbf{G}_\alpha$ must preserve every gain direction and the reference must provide two independent real projections of those directions. Closely spaced angles reduce $\sigma_{\min}(\mathbf{V})$ and hence both the field-domain and energy-domain stability margins.

### D. Finite-Alphabet Uniqueness and Noise Margin

For data detection, let $\mathcal{X}_{\mathrm{d}} = \mathcal{S}^{LK_{\mathrm{d}}}$. Since this set is finite, global uniqueness and a positive minimum separation are equivalent. Define
$$d^{(E)}_{\min,\mathrm{d}} \triangleq \min_{\substack{\mathbf{s}_1,\mathbf{s}_2\in\mathcal{X}_{\mathrm{d}}\\ \mathbf{s}_1\neq\mathbf{s}_2}} \left\| |\mathbf{G}_{\mathrm{d}}\mathbf{s}_1+\mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s}_2+\mathbf{b}_{\mathrm{d}}|^2 \right\|_2. \tag{72}$$

**Corollary 1** (Discrete uniqueness and stability)**.** *The symbols are uniquely identifiable if and only if $d^{(E)}_{\min,\mathrm{d}} > 0$. Equivalently, for every distinct pair,*
$$\mathbf{J}_{\mathrm{d}}\left(\frac{\mathbf{s}_1+\mathbf{s}_2}{2}\right)(\mathbf{s}_1-\mathbf{s}_2)_{\mathbb{R}} \neq \mathbf{0}. \tag{73}$$
*Moreover, the nearest-intensity detector is guaranteed to return the transmitted symbol vector whenever*
$$\|\mathbf{w}\|_2 < \frac{d^{(E)}_{\min,\mathrm{d}}}{2}. \tag{74}$$

For additive white Gaussian post-detection noise with covariance $\sigma_w^2\mathbf{I}$, the pairwise ML error probability is
$$P(\mathbf{s}_1 \to \mathbf{s}_2) = Q\left(\frac{d_E(\mathbf{s}_1,\mathbf{s}_2)}{2\sigma_w}\right), \tag{75}$$
so (72) is both a uniqueness certificate and the worst-case noise margin.

The data-phase field matrix has the factorization
$$\mathbf{G}_{\mathrm{d}} = (\mathbf{I}_{MK_{\mathrm{d}}} \otimes \widehat{\mathbf{H}})\boldsymbol{\Pi}\mathbf{P}_{\mathrm{tx},d}. \tag{76}$$

When the factors have full column rank,

$$\sigma_{\min}(\mathbf{G}_{\mathrm{d}}) \geq \sigma_{\min}(\widehat{\mathbf{H}})\sigma_{\min}(\mathbf{P}_{\mathrm{tx},d}), \tag{77}$$

$$\sigma_{\min}(\widehat{\mathbf{H}}) \approx \sigma_{\min}(\mathbf{V}\boldsymbol{\Gamma}) \geq \sigma_{\min}(\mathbf{V}) \min_{l} |\alpha_l|. \tag{78}$$

These inequalities quantify only preservation in the complex field domain. Theorem 1 adds the missing condition: the reference-induced real projection must also be nonzero for every constellation secant.

### *E. A Constructive Four-Phase Reference Bound*

The preceding conditions are exact but depend on the unknown midpoint. A phase-complete reference pattern gives a simple midpoint-independent sufficient condition. Suppose a base field sample $u_q = [\mathbf{G}\boldsymbol{\theta}]_q$ is repeated with four known references

$$b_{q,k} = \rho_q e^{j\phi_k}, \qquad \phi_k \in \{0, \pi/2, \pi, 3\pi/2\}, \quad \rho_q > 0. \tag{79}$$

Denote the resulting stacked mapping by $\mathcal{T}_{\mathrm{4ph}}(\boldsymbol{\theta})$.

**Theorem 2** (Four-phase global lower bound)**.** *For any* $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2$*,*

$$\begin{aligned}&\|\mathcal{T}_{\mathrm{4ph}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathrm{4ph}}(\boldsymbol{\theta}_2)\|_2^2 \\ &\quad = 4\left\||\mathbf{G}\boldsymbol{\theta}_1|^2 - |\mathbf{G}\boldsymbol{\theta}_2|^2\right\|_2^2 + 8\sum_q \rho_q^2 |[\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)]_q|^2.\end{aligned} \tag{80}$$

*Consequently, with* $\rho_{\min} = \min_q \rho_q$*,*

$$\|\mathcal{T}_{\mathrm{4ph}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathrm{4ph}}(\boldsymbol{\theta}_2)\|_2 \geq 2\sqrt{2}\rho_{\min}\|\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\|_2. \tag{81}$$

*If* $\mathbf{G}$ *has full column rank, then*

$$\alpha_{\mathcal{X}} \geq 2\sqrt{2}\rho_{\min}\sigma_{\min}(\mathbf{G}) > 0 \tag{82}$$

*for every feasible set* $\mathcal{X}$*. For a finite alphabet, full column rank can be replaced by* $\mathrm{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} = \varnothing$*.*

*Proof.* For one field coordinate, put $c_q = |u_{1,q}|^2 - |u_{2,q}|^2$ and $\Delta u_q = u_{1,q} - u_{2,q}$. Summing $[c_q + 2\rho_q \operatorname{Re}\{e^{-j\phi_k}\Delta u_q\}]^2$ over the four phases cancels all cross terms and gives $4c_q^2 + 8\rho_q^2|\Delta u_q|^2$. Summing over $q$ proves (80); the remaining claims follow from singular value or restricted-difference bounds. □

The construction also has a transparent holographic interpretation. Opposite reference phases cancel the unknown self-intensity:

$$\operatorname{Re}\{u_q\} = \frac{z_{q,0} - z_{q,\pi}}{4\rho_q}, \quad \operatorname{Im}\{u_q\} = \frac{z_{q,\pi/2} - z_{q,3\pi/2}}{4\rho_q}. \tag{83}$$

Thus, four phase-shifted energy measurements recover two orthogonal real projections of each complex field sample. This proves explicitly that reference phase diversity can remove the global phase ambiguity and yields the stability lower bound in (82). Fewer or hardware-constrained phases may still be unique, but their quality should be evaluated through $\alpha_{\mathcal{X}}$, $\underline{\sigma}_J$, or $d_{\min,\mathrm{d}}^{(E)}$, rather than through the rank of $\mathbf{G}$ alone.

The resulting design criteria are therefore

$$\max_{\mathbf{b}_{\mathrm{p}} \in \mathcal{B}_{\mathrm{p}}} \inf_{\boldsymbol{\theta} \in \mathcal{X}_{\mathrm{p}}} \sigma_{\min}[\mathbf{J}_{\mathrm{p}}(\boldsymbol{\theta})] \quad \text{and} \quad \max_{\mathbf{b}_{\mathrm{d}} \in \mathcal{B}_{\mathrm{d}}} d_{\min,\mathrm{d}}^{(E)} \tag{84}$$

for continuous training and discrete data detection, respectively. The first maximizes a uniform local/global lower bound, whereas the second directly maximizes the finite-alphabet decision margin.

## IV. Estimation and Detection Algorithms

Based on the uniqueness and stability analysis in Section III, this section develops practical recovery algorithms for the proposed holographic RIS receiver. The key point is that the algorithms should not be chosen only according to the algebraic rank of the equivalent matrix. Instead, the training stage should operate in a region where the intensity-domain Jacobian is well conditioned, while the data detector should preserve a positive energy-domain distance between finite-alphabet candidates.

The resulting receiver follows a two-stage structure. In the training stage, the unknown variables are the DOAs and the complex path gains. They are continuous parameters, and the far-field model provides a differentiable low-dimensional representation. A damped Gauss–Newton refinement is therefore used after a coarse initialization. In the data stage, the channel estimate is fixed and the unknown vector belongs to a finite constellation. Maximum-likelihood (ML) detection gives the finite-alphabet benchmark associated with the energy-domain minimum distance, whereas a reference-assisted Gerchberg–Saxton (GS) detector provides a lower-complexity iterative implementation for repeated data blocks.

### *A. Overall Two-Stage Recovery Framework*

The proposed receiver operates in two stages: a training stage for joint DOA and channel-gain estimation and a data stage for symbol detection.

During the training stage, the users transmit known pilot vectors $\mathbf{x}_{\mathrm{p}}[t] \in \mathbb{C}^L$, $t = 0, 1, \ldots, T_{\mathrm{p}} - 1$. The far-field structure in (24) is reused here by writing

$$\mathbf{H}(\boldsymbol{\eta}) = \mathbf{V}(\boldsymbol{\theta}, \boldsymbol{\varphi})\boldsymbol{\Gamma}(\boldsymbol{\alpha}), \tag{85}$$

where $\mathbf{V}(\boldsymbol{\theta}, \boldsymbol{\varphi})$ is the RIS array manifold determined by the users' DOAs, and $\boldsymbol{\Gamma}(\boldsymbol{\alpha}) = \mathrm{diag}(\alpha_1, \ldots, \alpha_L)$ collects the complex path gains. Thus, the training algorithm only needs to estimate the physical parameters that determine the manifold and gains, rather than all entries of $\mathbf{H}$ independently. We collect these unknowns in the real-valued vector

$$\boldsymbol{\eta} = \begin{bmatrix} \boldsymbol{\theta}^T & \boldsymbol{\varphi}^T & \operatorname{Re}\{\boldsymbol{\alpha}\}^T & \operatorname{Im}\{\boldsymbol{\alpha}\}^T \end{bmatrix}^T \in \mathbb{R}^{D_\eta}, \tag{86}$$

where $\boldsymbol{\theta}$ and $\boldsymbol{\varphi}$ represent the two angular coordinates, while $\operatorname{Re}\{\boldsymbol{\alpha}\}$ and $\operatorname{Im}\{\boldsymbol{\alpha}\}$ represent the complex gains in real form; hence $D_\eta = 4L$ in the two-dimensional DOA case. If only one angular parameter is considered for each user, the corresponding angular block is removed and the parameter dimension is reduced accordingly.

Define the stacked noiseless received field during the training stage as

$$\mathbf{g}_{\mathrm{p}}(\boldsymbol{\eta}) = \begin{bmatrix} \mathbf{H}(\boldsymbol{\eta})\mathbf{x}_{\mathrm{p}}[0] \\ \mathbf{H}(\boldsymbol{\eta})\mathbf{x}_{\mathrm{p}}[1] \\ \vdots \\ \mathbf{H}(\boldsymbol{\eta})\mathbf{x}_{\mathrm{p}}[T_{\mathrm{p}} - 1] \end{bmatrix} \in \mathbb{C}^{Q_{\mathrm{p}}}, \tag{87}$$

where $Q_{\mathrm{p}} = N_{\mathrm{r}} T_{\mathrm{p}}$ is the total number of training-stage intensity observations. The stacked training model is

$$\mathbf{z}_{\mathrm{p}} = |\mathbf{g}_{\mathrm{p}}(\boldsymbol{\eta}) + \mathbf{b}_{\mathrm{p}}|^2 + \mathbf{w}_{\mathrm{p}}, \tag{88}$$

**Algorithm 1** Two-Stage Reference-Assisted Holographic Reception

**Require:** Pilot vectors $\mathbf{x}_{\rm p}[t]$, training observations $\mathbf{z}_{\rm p}$, data observations $\mathbf{z}_{\rm d}$, RIS geometry, reference vectors $\mathbf{b}_{\rm p}$ and $\mathbf{b}_{\rm d}$, and constellation set $\mathcal{S}$.
**Ensure:** Estimated DOAs, estimated channel $\hat{\mathbf{H}}$, and detected symbols $\hat{\mathbf{s}}_{\rm d}$.
1: **Training stage:**
2: Construct the parametric field model $\mathbf{g}_{\rm p}(\boldsymbol{\eta})$ from the pilot vectors and RIS array manifold.
3: Obtain a coarse initial estimate $\boldsymbol{\eta}^{(0)}$.
4: Refine the DOAs and complex channel gains using the damped Gauss–Newton algorithm.
5: Reconstruct $\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}})$.
6: **Data stage:**
7: Construct $\hat{\mathbf{G}}_{\rm d} = \mathbf{G}_{\rm d}(\hat{\mathbf{H}})$.
8: Detect $\mathbf{s}_{\rm d}$ using maximum-likelihood detection or the proposed reference-assisted Gerchberg–Saxton detector.
9: Output the estimated DOAs, $\hat{\mathbf{H}}$, and $\hat{\mathbf{s}}_{\rm d}$.

where $\mathbf{b}_{\rm p} \in \mathbb{C}^{Q_{\rm p}}$ is the stacked known reference vector.

After estimating $\boldsymbol{\eta}$, the channel estimate is reconstructed as

$$\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}}). \tag{89}$$

During the data stage, the estimated channel is incorporated into the equivalent data-phase measurement matrix

$$\hat{\mathbf{G}}_{\rm d} = \mathbf{G}_{\rm d}(\hat{\mathbf{H}}). \tag{90}$$

Then according to (47) the data observation is modeled as

$$\mathbf{z}_{\rm d} = \left|\hat{\mathbf{G}}_{\rm d}\mathbf{s}_{\rm d} + \mathbf{b}_{\rm d}\right|^2 + \mathbf{w}_{\rm d}. \tag{91}$$

The stability results in Section III lead to three algorithmic design rules. First, the pilot matrix and RIS observation diversity must make the field response sensitive to different channel parameters; otherwise no nonlinear solver can restore the lost information. Second, the training reference should make the real Jacobian $\mathbf{J}_{\rm p}(\boldsymbol{\eta})$ well conditioned over the feasible angular region. Third, the data-phase reference should maximize the estimated minimum energy-domain distance between constellation vectors, since this distance is the decision margin of the ML detector and a useful indicator for the iterative detectors. The overall receiver is summarized in Algorithm 1.

The two stages are therefore connected through the estimated channel but employ different recovery mechanisms. The Gauss–Newton estimator exploits the differentiable low-dimensional channel parameterization, whereas the reference-assisted Gerchberg–Saxton detector exploits repeated projections and the finite-alphabet structure of communication symbols.

### B. Joint DOA and Channel-Gain Estimation

*1) Nonlinear Least-Squares Formulation:* Based on the block-level pilot observation model in (45), the predicted noiseless intensity corresponding to a candidate parameter vector $\boldsymbol{\eta}$ is defined as

$$\hat{\mathbf{z}}_{\rm p}(\boldsymbol{\eta}) = \left|\mathbf{g}_{\rm p}(\boldsymbol{\eta}) + \mathbf{b}_{\rm p}\right|^2. \tag{92}$$

The joint DOA and channel-gain estimation problem is formulated as

$$\hat{\boldsymbol{\eta}} = \arg\min_{\boldsymbol{\eta}\in\Omega} \frac{1}{2}\left\|\mathbf{z}_{\rm p} - \left|\mathbf{g}_{\rm p}(\boldsymbol{\eta}) + \mathbf{b}_{\rm p}\right|^2\right\|_2^2, \tag{93}$$

where $\Omega$ denotes the feasible parameter region.

This formulation directly exploits the far-field array structure. Compared with estimating all entries of $\mathbf{H} \in \mathbb{C}^{N_{\rm r}\times L}$ independently, the parametric model substantially reduces the number of unknowns when the number of RIS elements is large. More importantly, the optimization variable is exactly the physical quantity that determines the array manifold. Hence, the Jacobian used by the numerical solver also has a direct identifiability interpretation: poorly separated DOAs or weak path gains appear as small singular values of the intensity-domain Jacobian.

*2) Jacobian of the Parametric Intensity Mapping:* At the $t$-th pilot instant, the noise-free received field is given by

$$\mathbf{g}_{\rm p}[t] = \sum_{l=1}^{L} \alpha_l x_{{\rm p},l}[t]\mathbf{v}_l, \tag{94}$$

where $x_{{\rm p},l}[t]$ denotes the pilot symbol transmitted by the $l$-th user, corresponding to the $l$-th entry of the pilot vector $\mathbf{x}_{\rm p}[t]$.

Let $\boldsymbol{\psi}_l \in \mathbb{R}^{N_{\rm r}}$ denote the phase vector whose $\iota(n,m)$-th entry is $[\boldsymbol{\psi}_l]_{\iota(n,m)} = \psi_{n,m}^{(l)}$. Since $\mathbf{v}_l(\theta_l,\phi_l) = \exp(j\boldsymbol{\psi}_l)$, where the exponential is applied element-wise, its angular derivatives can be written as

$$\frac{\partial \mathbf{v}_l}{\partial \theta_l} = j\frac{\partial \boldsymbol{\psi}_l}{\partial \theta_l} \odot \mathbf{v}_l, \quad \frac{\partial \mathbf{v}_l}{\partial \phi_l} = j\frac{\partial \boldsymbol{\psi}_l}{\partial \phi_l} \odot \mathbf{v}_l, \tag{95}$$

where the $\iota(n,m)$-th entries are $[\partial\boldsymbol{\psi}_l/\partial\theta_l]_{\iota(n,m)} = k_0\cos\theta_l\,(y_n\cos\phi_l + z_m\sin\phi_l)$ and $[\partial\boldsymbol{\psi}_l/\partial\phi_l]_{\iota(n,m)} = k_0\sin\theta_l\,(-y_n\sin\phi_l + z_m\cos\phi_l)$.

Using (94), and defining $\alpha_{l,R} = \mathrm{Re}\{\alpha_l\}$ and $\alpha_{l,I} = \mathrm{Im}\{\alpha_l\}$, the field derivatives at pilot instant $t$ are

$$\begin{aligned} \frac{\partial \mathbf{g}_{\rm p}[t]}{\partial \theta_l} &= \alpha_l x_{{\rm p},l}[t]\frac{\partial \mathbf{v}_l}{\partial \theta_l}, & \frac{\partial \mathbf{g}_{\rm p}[t]}{\partial \phi_l} &= \alpha_l x_{{\rm p},l}[t]\frac{\partial \mathbf{v}_l}{\partial \phi_l}, \\ \frac{\partial \mathbf{g}_{\rm p}[t]}{\partial \alpha_{l,R}} &= x_{{\rm p},l}[t]\mathbf{v}_l, & \frac{\partial \mathbf{g}_{\rm p}[t]}{\partial \alpha_{l,I}} &= j x_{{\rm p},l}[t]\mathbf{v}_l. \end{aligned} \tag{96}$$

The derivatives over all pilot instants are stacked in the same order as $\mathbf{g}_{\rm p}(\boldsymbol{\eta})$. Thus, the Jacobian of the parametric intensity mapping is

$$\mathbf{J}_{\rm p}(\boldsymbol{\eta}) = 2\,\mathrm{Re}\left\{\mathrm{diag}\big(\mathbf{u}_{\rm p}^*(\boldsymbol{\eta})\big)\,\frac{\partial \mathbf{g}_{\rm p}(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^{\mathsf{T}}}\right\} \in \mathbb{R}^{Q_{\rm p}\times D_\eta}. \tag{97}$$

The same Jacobian determines both the local identifiability discussed in Section III and the numerical behavior of the Gauss–Newton iterations. In particular, a small $\sigma_{\min}(\mathbf{J}_{\rm p})$ indicates that certain perturbations in the DOAs or path gains produce only weak intensity variations, leading to both poor estimation stability and poorly conditioned Gauss–Newton updates. This is the reason for using phase-diverse references during training: the reference term changes the real projection directions in the Jacobian and can convert a nearly invisible complex-field perturbation into a measurable intensity perturbation.

*3) Damped Gauss–Newton Update:* At iteration $i$, the residual is locally approximated as

$$\mathbf{e}_{\mathrm{p}}\left(\boldsymbol{\eta}^{(i)}+\Delta\boldsymbol{\eta}\right)\approx\mathbf{e}_{\mathrm{p}}^{(i)}-\mathbf{J}_{\mathrm{p}}^{(i)}\Delta\boldsymbol{\eta}, \tag{98}$$

where $\mathbf{u}_{\mathrm{p}}^{(i)}=\mathbf{g}_{\mathrm{p}}\left(\boldsymbol{\eta}^{(i)}\right)+\mathbf{b}_{\mathrm{p}}$, $\mathbf{e}_{\mathrm{p}}^{(i)}=\mathbf{z}_{\mathrm{p}}-\left|\mathbf{u}_{\mathrm{p}}^{(i)}\right|^2$, and $\mathbf{J}_{\mathrm{p}}^{(i)}=\mathbf{J}_{\mathrm{p}}(\boldsymbol{\eta}^{(i)})$ denotes the intensity-domain Jacobian evaluated at the current iterate.

The damped Gauss–Newton increment is then computed as

$$\Delta\boldsymbol{\eta}^{(i)}=\left[\left(\mathbf{J}_{\mathrm{p}}^{(i)}\right)^T\mathbf{J}_{\mathrm{p}}^{(i)}+\lambda^{(i)}\mathbf{I}\right]^{-1}\left(\mathbf{J}_{\mathrm{p}}^{(i)}\right)^T\mathbf{e}_{\mathrm{p}}^{(i)}, \tag{99}$$

where $\lambda^{(i)}\geq 0$ is a damping parameter introduced to improve numerical stability when the Jacobian is poorly conditioned. The parameter vector is updated as

$$\boldsymbol{\eta}^{(i+1)}=\Pi_{\Omega}\left(\boldsymbol{\eta}^{(i)}+\Delta\boldsymbol{\eta}^{(i)}\right), \tag{100}$$

where $\Pi_{\Omega}(\cdot)$ projects the angular parameters onto the feasible region. In practice, the damping parameter is decreased when the update reduces the objective function and increased otherwise.

*4) Initialization:* Since the nonlinear least-squares problem is nonconvex, a suitable initialization is important for the subsequent Gauss–Newton refinement. The initialization is designed to first obtain a coarse estimate of the complex field or channel and then fit the far-field parametric model.

When the four-phase reference pattern in Theorem 2 is available during training, the complex field can be directly initialized from opposite reference phases. For a fixed pilot time and RIS element, let $z_{q,0}$, $z_{q,\pi/2}$, $z_{q,\pi}$, and $z_{q,3\pi/2}$ denote the four intensity measurements with reference amplitude $\rho_q$. The corresponding field sample is estimated as

$$\hat{u}_q=\frac{z_{q,0}-z_{q,\pi}}{4\rho_q}+j\frac{z_{q,\pi/2}-z_{q,3\pi/2}}{4\rho_q}. \tag{101}$$

Stacking the recovered field samples gives $\hat{\mathbf{g}}_{\mathrm{p}}$. A coarse unstructured channel estimate is then obtained by least squares as

$$\mathbf{h}^{(0)}=\mathbf{G}_{\mathrm{p}}^{\dagger}\hat{\mathbf{g}}_{\mathrm{p}},\qquad\mathbf{H}^{(0)}=\mathrm{unvec}\left(\mathbf{h}^{(0)}\right). \tag{102}$$

When four separate reference phases are unavailable, the reference-dominated approximation can be used to obtain

$$\mathbf{h}^{(0)}=\arg\min_{\mathbf{h}}\left\|\mathbf{z}_{\mathrm{p}}-|\mathbf{b}_{\mathrm{p}}|^2-2\,\mathrm{Re}\left\{\mathbf{b}_{\mathrm{p}}^{*}\odot\mathbf{G}_{\mathrm{p}}\mathbf{h}\right\}\right\|_2^2, \tag{103}$$

from which $\mathbf{H}^{(0)}$ is reconstructed.

The initial DOA of each user is then obtained through a coarse angular search,

$$\left(\theta_l^{(0)},\varphi_l^{(0)}\right)=\arg\max_{(\theta,\varphi)\in\mathcal{G}_{\Omega}}\frac{\left|\mathbf{v}^H(\theta,\varphi)\mathbf{h}_l^{(0)}\right|^2}{\|\mathbf{v}(\theta,\varphi)\|_2^2}, \tag{104}$$

where $\mathcal{G}_{\Omega}$ denotes a coarse angular grid. Given the coarse DOA estimate, the corresponding complex path gain is obtained by least-squares projection onto the steering vector. These coarse parameter estimates are then used to initialize the damped Gauss–Newton iterations. The complete joint estimation procedure is summarized in Algorithm 2.

**Algorithm 2** Joint DOA and Channel-Gain Estimation

**Require:** Training observations $\mathbf{z}_{\mathrm{p}}$, pilots $\mathbf{x}_{\mathrm{p}}[t]$, reference vector $\mathbf{b}_{\mathrm{p}}$, RIS geometry, feasible parameter region $\Omega$, maximum iteration number $I_{\mathrm{c}}$, initial damping factor $\lambda^{(0)}$, and threshold $\epsilon_{\mathrm{c}}$.
**Ensure:** Estimated parameter vector $\hat{\boldsymbol{\eta}}$ and channel $\hat{\mathbf{H}}$.

```
Obtain H^(0) using (103).
Obtain coarse DOAs using (104).
Form η^(0) according to (86).
for i = 0, 1, ..., I_c − 1 do
    Compute u_p^(i) = g_p(η^(i)) + b_p.
    Compute the residual e_p^(i) = z_p − |u_p^(i)|^2.
    Construct J_p^(i) using (97).
    Compute Δη^(i) using (99).
    Compute the projected trial point using (100).
    if the trial point decreases F_p then
        Accept η^(i+1) = η_trial^(i+1).
        Reduce the damping parameter.
    else
        Set η^(i+1) = η^(i).
        Increase the damping parameter.
    end if
    if ||Δη^(i)||_2 / (||η^(i)||_2 + ε_0) < ε_c then
        break
    end if
end for
Set η̂ = η^(i+1).
Reconstruct Ĥ = H(η̂).
```

The Gauss–Newton method is a local optimization procedure and does not guarantee convergence to the global minimizer from an arbitrary initialization. Its use is justified here by the differentiable low-dimensional channel parameterization, the availability of pilot observations, and the possibility of obtaining a coarse angular initialization before iterative refinement.

## C. Finite-Alphabet Symbol Detection

After the training stage, the estimated channel is used to construct $\hat{\mathbf{G}}_{\mathrm{d}}$. The symbol detection problem is based on

$$\mathbf{z}_{\mathrm{d}}=\left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_{\mathrm{d}}+\mathbf{b}_{\mathrm{d}}\right|^2+\mathbf{w}_{\mathrm{d}},\qquad\mathbf{s}_{\mathrm{d}}\in\mathcal{S}^{D_{\mathrm{s}}}. \tag{105}$$

Unlike the training-stage unknowns, the transmitted symbols belong to a finite constellation. Therefore, the relevant theoretical quantity is the energy-domain separation between candidate symbol vectors rather than the full-rank condition of a continuous Jacobian. A detector can be reliable only when the estimated mapping assigns distinct and sufficiently separated intensity vectors to different constellation points. This motivates using ML detection as a benchmark and using the same distance metric to initialize and evaluate low-complexity iterative detectors.

*1) Maximum-Likelihood Detection:* Assuming independent Gaussian post-detection noise with identical variance, the maximum-likelihood detector is equivalent to

$$\hat{\mathbf{s}}_{\mathrm{ML}} = \arg\min_{\mathbf{s}\in\mathcal{S}^{D_{\mathrm{s}}}} \left\| \mathbf{z}_{\mathrm{d}} - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}\right|^2 \right\|_2^2. \tag{106}$$

For two candidates $\mathbf{s}_i$ and $\mathbf{s}_j$, define

$$\hat{d}_E\big(\mathbf{s}_i, \mathbf{s}_j\big) = \left\| \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_i + \mathbf{b}_{\mathrm{d}}\right|^2 - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_j + \mathbf{b}_{\mathrm{d}}\right|^2 \right\|_2. \tag{107}$$

The corresponding minimum energy-domain distance is

$$\hat{d}^{(E)}_{\mathrm{min,d}} = \min_{\mathbf{s}_i\neq\mathbf{s}_j} \hat{d}_E\big(\mathbf{s}_i, \mathbf{s}_j\big). \tag{108}$$

The ML detector provides a direct connection between the identifiability theory in Section III and the simulated symbol error rate. In particular, a positive $\hat{d}^{(E)}_{\mathrm{min,d}}$ ensures that different candidate symbol vectors produce different noiseless intensity observations under the estimated model.

However, exhaustive ML detection requires evaluating $|\mathcal{S}|^{D_{\mathrm{s}}}$ candidate vectors. Its complexity grows exponentially with the detected block length, making it primarily suitable for short blocks and performance benchmarking.

*2) Linearized Initialization and Candidate Screening:* The reference term also provides a simple way to initialize the data detector. If the reference-induced interference term dominates the unknown self-intensity over the current block, then

$$\mathbf{z}_{\mathrm{d}} - |\mathbf{b}_{\mathrm{d}}|^2 \approx 2\,\mathrm{Re}\{\mathbf{b}_{\mathrm{d}}^* \odot \hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_{\mathrm{d}}\}. \tag{109}$$

With $\bar{\mathbf{s}} = [\mathrm{Re}\{\mathbf{s}\}^T, \mathrm{Im}\{\mathbf{s}\}^T]^T$, define the real-valued linearized data matrix

$$\mathbf{A}_{\mathrm{d}} = 2\left[\mathrm{Re}\{\mathrm{diag}(\mathbf{b}_{\mathrm{d}}^*)\hat{\mathbf{G}}_{\mathrm{d}}\} \quad -\mathrm{Im}\{\mathrm{diag}(\mathbf{b}_{\mathrm{d}}^*)\hat{\mathbf{G}}_{\mathrm{d}}\}\right]. \tag{110}$$

A regularized least-squares estimate is first obtained in the real domain as

$$\bar{\mathbf{s}}_{\mathrm{lin}} = \left(\mathbf{A}_{\mathrm{d}}^T\mathbf{A}_{\mathrm{d}} + \gamma_{\mathrm{lin}}\mathbf{I}\right)^{-1}\mathbf{A}_{\mathrm{d}}^T\left(\mathbf{z}_{\mathrm{d}} - |\mathbf{b}_{\mathrm{d}}|^2\right), \tag{111}$$

where $\gamma_{\mathrm{lin}} \geq 0$ is a small regularization parameter. The corresponding complex estimate is reconstructed as $\mathbf{s}_{\mathrm{lin,cont}} = [\mathbf{I}; j\mathbf{I}]\bar{\mathbf{s}}_{\mathrm{lin}}$, followed by element-wise constellation projection $\mathbf{s}_{\mathrm{lin}} = \mathcal{Q}_{\mathcal{S}}(\mathbf{s}_{\mathrm{lin,cont}})$.

This vector is used either as the initial point of an iterative detector or as the center of a reduced candidate list. For example, one may enumerate only the constellation vectors whose Hamming distance from $\mathbf{s}_{\mathrm{lin}}$ is no larger than a small integer. This screened ML detector is no longer globally optimal, but it often captures most of the ML gain when $\hat{d}^{(E)}_{\mathrm{min,d}}$ is not too small.

*3) Reference-Assisted Gerchberg–Saxton Detection:* The conventional Gerchberg–Saxton algorithm alternates between a measurement-magnitude constraint and a signal-domain constraint. The detector considered here is not a direct application of the conventional homogeneous phase-retrieval algorithm. Instead, it is adapted to the affine observation model by including the known reference vector in the forward projection and subtracting it before the signal-domain back-projection.

Given the current symbol estimate $\mathbf{s}^{(i)}$, the predicted affine field is

$$\mathbf{u}_{\mathrm{d}}^{(i)} = \hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}^{(i)} + \mathbf{b}_{\mathrm{d}}. \tag{112}$$

**Algorithm 3** Reference-Assisted Gerchberg–Saxton Symbol Detection

**Require:** Data observations $\mathbf{z}_{\mathrm{d}}$, matrix $\hat{\mathbf{G}}_{\mathrm{d}}$, reference vector $\mathbf{b}_{\mathrm{d}}$, constellation $\mathcal{S}$, regularization factor $\gamma$, maximum iteration number $I_{\mathrm{d}}$, and threshold $\epsilon_{\mathrm{d}}$.
**Ensure:** Detected symbol vector $\hat{\mathbf{s}}_{\mathrm{d}}$.

Compute $\mathbf{r}_{\mathrm{d}} = \sqrt{[\mathbf{z}_{\mathrm{d}}]_+}$.
Precompute $\mathbf{B}_{\mathrm{d}}$ according to (114).
Initialize $\mathbf{s}^{(0)}$ using a linearized estimate, the previous data block, or a random constellation vector.
**for** $i = 0, 1, \ldots, I_{\mathrm{d}} - 1$ **do**
  Compute the predicted affine field using (112).
  Apply the magnitude constraint using (113).
  Back-project using (115).
  Apply the constellation projection using (116).
  **if** $\|\mathbf{s}^{(i+1)} - \mathbf{s}^{(i)}\|_2 / \big(\|\mathbf{s}^{(i)}\|_2 + \epsilon_0\big) < \epsilon_{\mathrm{d}}$ **then**
    **break**
  **end if**
**end for**
Output $\hat{\mathbf{s}}_{\mathrm{d}} = \mathcal{Q}_{\mathcal{S}}\big(\mathbf{s}^{(i+1)}\big)$.

The measured-magnitude constraint is then enforced as

$$\tilde{\mathbf{u}}_{\mathrm{d}}^{(i)} = \sqrt{[\mathbf{z}_{\mathrm{d}}]_+} \odot \exp\Big(j\angle\mathbf{u}_{\mathrm{d}}^{(i)}\Big), \tag{113}$$

where $[\mathbf{z}]_+ = \max(\mathbf{z}, \mathbf{0})$ is applied element-wise.

The reference vector is then removed, and the resulting field is projected back onto the signal domain. To improve numerical stability when $\hat{\mathbf{G}}_{\mathrm{d}}$ is ill-conditioned, define the regularized back-projection matrix

$$\mathbf{B}_{\mathrm{d}} = \big(\hat{\mathbf{G}}_{\mathrm{d}}^{\mathsf{H}}\hat{\mathbf{G}}_{\mathrm{d}} + \gamma\mathbf{I}\big)^{-1}\hat{\mathbf{G}}_{\mathrm{d}}^{\mathsf{H}}, \tag{114}$$

where $\gamma \geq 0$ is a regularization parameter. When $\gamma = 0$ and $\hat{\mathbf{G}}_{\mathrm{d}}$ has full column rank, $\mathbf{B}_{\mathrm{d}} = \hat{\mathbf{G}}_{\mathrm{d}}^{\dagger}$.

The continuous signal-domain update is

$$\tilde{\mathbf{s}}^{(i+1)} = \mathbf{B}_{\mathrm{d}}\big(\tilde{\mathbf{u}}_{\mathrm{d}}^{(i)} - \mathbf{b}_{\mathrm{d}}\big). \tag{115}$$

The finite-alphabet structure is incorporated through

$$\mathbf{s}^{(i+1)} = \mathcal{Q}_{\mathcal{S}}\big(\tilde{\mathbf{s}}^{(i+1)}\big), \tag{116}$$

where $\mathcal{Q}_{\mathcal{S}}(\cdot)$ denotes element-wise nearest-neighbor quantization onto the constellation $\mathcal{S}$.

The projection in (116) may be applied after every iteration or only after the continuous iterations converge. Applying it at every iteration exploits the finite-alphabet prior more strongly but may cause premature stagnation when the current estimate is far from the transmitted symbol. Applying it only at the final iteration generally produces a smoother continuous iteration trajectory. The complete detector is summarized in Algorithm 3.

The proposed reference-assisted GS detector is attractive for repeated data detection because $\mathbf{B}_{\mathrm{d}}$ can be precomputed whenever the estimated channel remains unchanged over a coherence block. Each subsequent iteration then mainly involves matrix-vector multiplications and element-wise operations.

### D. Computational Complexity and Implementation Considerations

Let $I_{\mathrm{c}}$ and $I_{\mathrm{d}}$ denote the numbers of channel-estimation and symbol-detection iterations, respectively. The principal computational costs are summarized in Table I.

The cost in Table I separates operations that are repeated every iteration from operations that can be amortized over a coherence block. For joint DOA and gain estimation, the dominant operations are the formation of $(\mathbf{J}_{\mathrm{p}})^{\mathsf{T}}\mathbf{J}_{\mathrm{p}}$ and the solution of the $D_\eta \times D_\eta$ linear system. Because $D_\eta = 4L$ for the two-dimensional far-field model, the parameter dimension does not increase directly with the number of RIS elements. This represents a substantial reduction compared with unstructured full-channel estimation, whose real-valued dimension is $2N_{\mathrm{r}}L$.

For the GS detector, the regularized back-projection matrix in (114) requires a one-time cost of approximately $\mathcal{O}\left(Q_{\mathrm{d}}D_{\mathrm{s}}^2 + D_{\mathrm{s}}^3\right)$. This cost is amortized over all data blocks transmitted within the same channel-coherence interval. Once $\mathbf{B}_{\mathrm{d}}$ has been precomputed, each iteration requires minimal computational effort, primarily involving two matrix-vector multiplications and element-wise magnitude or phase operations.

The linearized detector in (111) has a similar one-time matrix-inversion cost when $\hat{\mathbf{G}}_{\mathrm{d}}$ and $\mathbf{b}_{\mathrm{d}}$ are fixed. Therefore, it is useful not only as a standalone low-complexity detector but also as a deterministic initial point for GS or projected GN iterations.

The data detector may also utilize warm-start initialization. If the channel and transmitted waveform vary slowly between adjacent blocks, the estimate from the previous block can serve as $\mathbf{s}^{(0)}$. For independently modulated blocks, a reference-dominated linearized estimate or multiple random constellation initializations may be employed.

The stopping conditions should include a small positive constant $\epsilon_0$ in the denominator to avoid numerical instability when the current estimate has a small norm. In addition to the relative-update criterion used in Algorithms 2 and 3, a residual-based condition can be employed:

$$\frac{\left\|\mathbf{z} - \left|\mathbf{G}\boldsymbol{\xi}^{(i)} + \mathbf{b}\right|^2\right\|_2}{\|\mathbf{z}\|_2 + \epsilon_0} < \epsilon_{\mathrm{r}}, \tag{117}$$

where $\boldsymbol{\xi}^{(i)}$ denotes the current channel-parameter or symbol estimate.

### E. Cramér–Rao Lower Bound for DOA Estimation

The CRLB follows directly from the parametric training model in (88) and its Jacobian in (97). Assume that the effective post-detection noise is white Gaussian, $\mathbf{w}_{\mathrm{p}} \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathrm{p}}^2\mathbf{I})$, with variance independent of the unknown parameters. Repartition the real parameter vector in (86) as $\boldsymbol{\eta} = [\boldsymbol{\psi}^{\mathsf{T}}, \boldsymbol{\nu}^{\mathsf{T}}]^{\mathsf{T}}$, where $\boldsymbol{\psi} = [\boldsymbol{\theta}^{\mathsf{T}}, \boldsymbol{\varphi}^{\mathsf{T}}]^{\mathsf{T}}$ contains the DOAs and $\boldsymbol{\nu} = [\mathrm{Re}\{\boldsymbol{\alpha}\}^{\mathsf{T}}, \mathrm{Im}\{\boldsymbol{\alpha}\}^{\mathsf{T}}]^{\mathsf{T}}$ contains the nuisance path gains. The Fisher information matrix (FIM) is then

$$\mathbf{F}(\boldsymbol{\eta}) = \frac{1}{\sigma_{\mathrm{p}}^2}\mathbf{J}_{\mathrm{p}}^{\mathsf{T}}(\boldsymbol{\eta})\mathbf{J}_{\mathrm{p}}(\boldsymbol{\eta}), \tag{118}$$

with the column partition $\mathbf{J}_{\mathrm{p}} = [\mathbf{J}_\psi \ \mathbf{J}_\nu]$. Eliminating the unknown gains gives the equivalent FIM for the DOAs,

$$\begin{aligned} \mathbf{F}_{\mathrm{DOA}} &= \mathbf{F}_{\psi\psi} - \mathbf{F}_{\psi\nu}\mathbf{F}_{\nu\nu}^{\dagger}\mathbf{F}_{\nu\psi} \\ &= \frac{1}{\sigma_{\mathrm{p}}^2}\mathbf{J}_\psi^{\mathsf{T}}\mathbf{P}_\nu^{\perp}\mathbf{J}_\psi, \\ \mathbf{P}_\nu^{\perp} &= \mathbf{I} - \mathbf{J}_\nu\mathbf{J}_\nu^{\dagger}, \end{aligned} \tag{119}$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose inverse. Thus, only the component of an angle-induced intensity perturbation orthogonal to the gain perturbation subspace carries DOA information. If $\mathbf{F}_{\mathrm{DOA}}$ is nonsingular, any unbiased DOA estimator satisfies

$$\mathrm{Cov}(\hat{\boldsymbol{\psi}}) \succeq \mathbf{F}_{\mathrm{DOA}}^{-1}. \tag{120}$$

In particular, for user $l$,

$$\mathrm{var}(\hat{\theta}_l) \geq \left[\mathbf{F}_{\mathrm{DOA}}^{-1}\right]_{\theta_l,\theta_l}, \qquad \mathrm{var}(\hat{\varphi}_l) \geq \left[\mathbf{F}_{\mathrm{DOA}}^{-1}\right]_{\varphi_l,\varphi_l}. \tag{121}$$

Equation (119) also sharpens the earlier identifiability interpretation: weak singular directions of the gain-projected Jacobian $\mathbf{P}_\nu^{\perp}\mathbf{J}_\psi$ produce large DOA bounds, while a rank deficiency makes at least one angular direction locally unidentifiable and its ordinary CRLB unbounded. Hence the pilot and reference designs that improve Jacobian conditioning also improve the attainable DOA accuracy.

## V. Simulation Results

This section presents numerical results to assess the performance of the proposed reference-assisted holographic RIS receiver. We first evaluate the channel estimation accuracy during the training phase, followed by the symbol detection performance under both perfect and estimated channel state information (CSI). We then investigate how the reference-wave design and RIS measurement diversity affect energy-domain separability and detection reliability.

Unless otherwise stated, the simulations follow the two-phase model in Section II with a two-user $8 \times 8$ holographic RIS operating at 3.5 GHz and half-wavelength element spacing. The complex path gains are $\alpha_1 = e^{j0.35}$ and $\alpha_2 = 0.85e^{-j0.55}$, normalized QPSK is used, and the reference amplitude is 1.5. Training and data observations follow (88) and (47) with independent real Gaussian post-detection noise of variance $\sigma_w^2$ per intensity sample. For a noiseless intensity vector $\boldsymbol{\mu} = |\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b}|^2 \in \mathbb{R}^Q$, the SNR is defined as

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\mathbb{E}\{\|\boldsymbol{\mu}\|_2^2\}}{Q\sigma_w^2}\right). \tag{122}$$

Scenario-specific DOAs, pilot lengths, modulation orders, and reference or RIS configurations are stated with the corresponding experiments.

### A. Channel Estimation Performance

We evaluate the training-stage estimator under the intensity-domain noise model specified above. Unless stated otherwise, two users, an $8 \times 8$ RIS, $T_{\mathrm{p}} = 32$ QPSK pilot observations, and a reference amplitude of 1.5 are used. The user DOAs are $(15^\circ, 8^\circ)$ and $(35^\circ, -10^\circ)$, and the path-gain magnitudes are 1 and 0.85, respectively. Each point is averaged over 30

TABLE I: Computational Complexity of the Main Recovery Algorithms

| Method | Unknown dimension | Approximate complexity | Primary role |
|---|---|---|---|
| Joint damped GN | $D_\eta$ | $\mathcal{O}\left(I_c\left(Q_p D_\eta^2 + D_\eta^3\right)\right)$ | DOA and channel-gain estimation |
| Exhaustive ML | $D_s$ | $\mathcal{O}\left(\lvert\mathcal{S}\rvert^{D_s} Q_d D_s\right)$ | Performance benchmark |
| Linearized detector | $D_s$ | $\mathcal{O}\left(Q_d D_s^2 + D_s^3\right)$ | Initialization and candidate screening |
| Reference-assisted GS | $D_s$ | $\mathcal{O}(I_d Q_d D_s)$ | Low-complexity symbol detection |

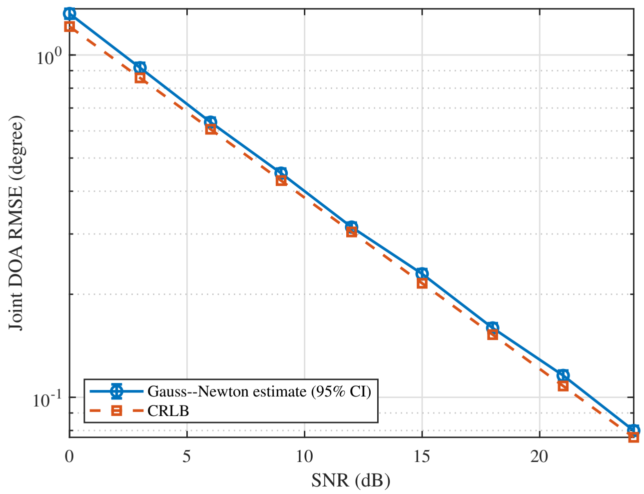


Fig. 2: Joint DOA RMSE and the corresponding CRLB versus SNR.

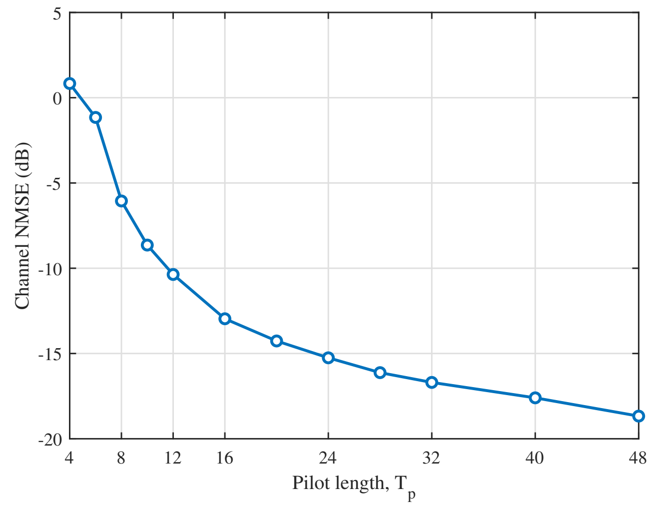


Fig. 3: Channel-estimation NMSE versus pilot length at an SNR of 12 dB.

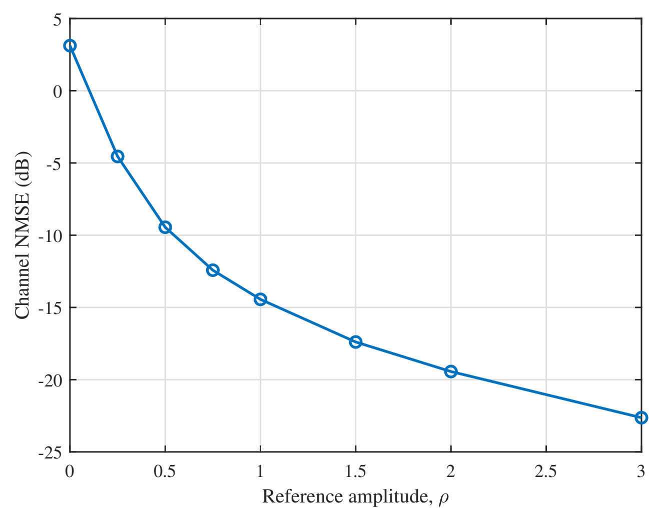


Fig. 4: Channel-estimation NMSE versus reference amplitude for fixed post-detection noise variance.

independent noise realizations unless noted otherwise. The reference phase pattern is selected from a 32-entry codebook by maximizing the minimum singular value of the training Jacobian.

To assess the angular accuracy of the recovered channel, the estimate is projected onto the continuous array manifold. Based on $1000$ independent noise realizations, Fig. 2 shows that the joint DOA RMSE decreases from $1.32°$ at 0 dB to $0.31°$ at 12 dB and $0.08°$ at 24 dB, closely following the slope and scale predicted by the CRLB. Because the implemented estimator first recovers the unstructured channel and then projects it onto the array manifold, it is generally biased at finite SNR; the ordinary unbiased-estimator CRLB is therefore used here as a local accuracy benchmark rather than a strict finite-sample bound for every operating point.

Fig. 3 examines the role of training length at 12 dB. With $T_p = 4$, which only meets the real-dimensional counting requirement for two complex user channels at each RIS element, the NMSE is $0.83$ dB. Increasing the pilot length to 12, 32, and 48 improves the NMSE to $-10.37$, $-16.70$, and $-18.67$ dB, respectively. Thus, satisfying the observation count alone is insufficient; additional pilot diversity improves Jacobian conditioning, although the gain gradually diminishes once the system is well overdetermined.

The reference-amplitude sweep in Fig. 4 keeps the post-detection noise variance fixed at its value for the nominal $\rho = 1.5$ operating point. Without a reference, the NMSE is $3.12$ dB; even a reference amplitude of $0.25$ reduces it to $-4.55$ dB, while $\rho = 1.5$ gives $-17.39$ dB. The continuing improvement at larger amplitudes reflects the stronger interference term in the ideal additive-noise model. It does not account for detector saturation or quantization, and hence should not be interpreted as favoring arbitrarily large reference power in hardware.

Finally, Fig. 5 compares reference phase patterns over a range of SNRs at equal reference power. Each point is averaged over $300$ paired Monte Carlo trials. Without a reference, the channel NMSE remains near 3 dB throughout the simulated range, consistent with the numerically singular training Jacobian ($\sigma_{\min} = 3.69 \times 10^{-16}$). In contrast, the constant, random, and uniform reference patterns yield minimum singular values of $8.64$, $8.25$, and $9.95$, respec-

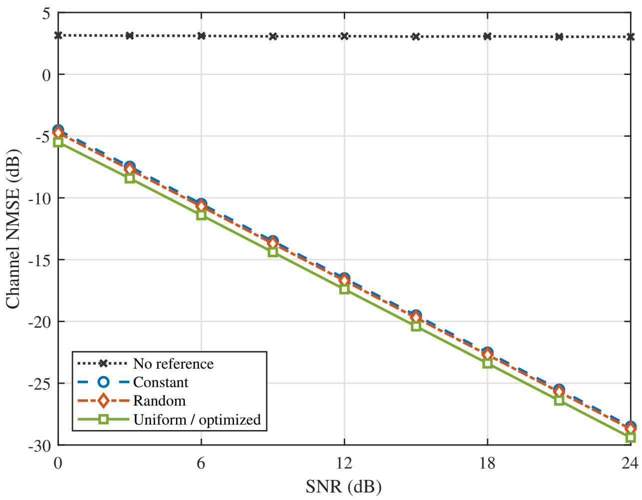


Fig. 5: Channel-estimation NMSE versus SNR for different training-reference designs.

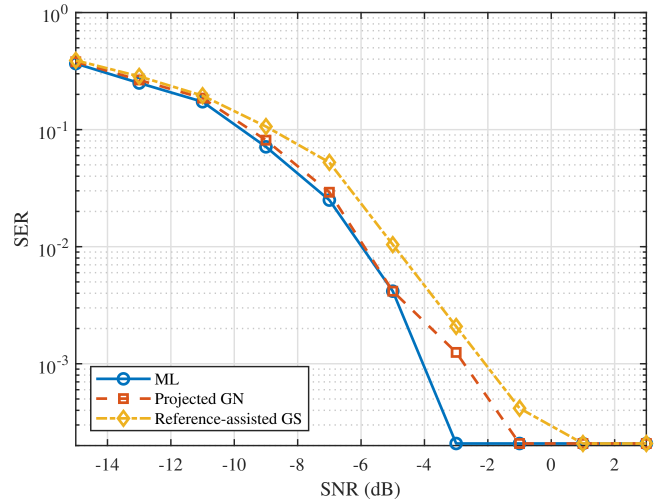


Fig. 6: SER of different symbol detectors under perfect CSI.

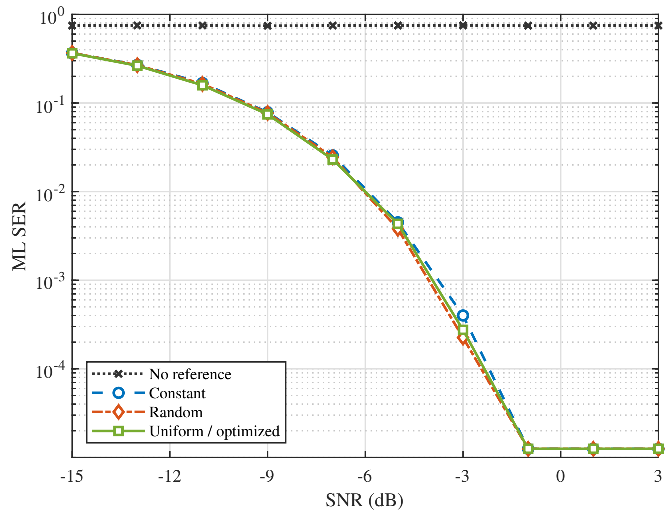


Fig. 7: ML SER versus SNR for different data-phase reference designs under perfect CSI. Zero-error points are displayed at the half-event level.

tively, and their NMSEs decrease steadily with SNR. From 0 to 24 dB, the corresponding NMSEs improve from $-4.52$, $-4.74$, and $-5.50$ dB to $-28.51$, $-28.71$, and $-29.40$ dB. The uniformly spaced pattern is also selected by the finite codebook search; hence, the uniform and optimized curves coincide. These results show that the reference removes the phase ambiguity responsible for the error floor, while improved Jacobian conditioning provides an additional accuracy gain.

### *B. Symbol Detection Performance with Perfect CSI*

We first isolate the data detector from channel-estimation errors by constructing $\mathbf{G}_{\mathrm{d}}$ from the true channel. Unless stated otherwise, two QPSK users at the DOAs used in the preceding subsection, four reference states, and a reference amplitude of $1.5$ are considered.

Fig. 6 compares the three detectors under the codebook-selected reference. ML gives the finite-alphabet benchmark, while projected GN remains close to it throughout the waterfall region. The reference-assisted GS detector has a moderate loss due to its alternating projections but preserves the same SNR trend. For example, at $-9$ dB the SERs of ML, projected GN, and reference-assisted GS are $7.13 \times 10^{-2}$, $8.13 \times 10^{-2}$, and $1.06 \times 10^{-1}$, respectively; at $-5$ dB they decrease to $4.17 \times 10^{-3}$, $4.17 \times 10^{-3}$, and $1.04 \times 10^{-2}$.

The role of the reference is shown in Fig. 7. Because the SNR in (122) is normalized by the noiseless intensity power, reference designs of different field patterns can be compared at the same effective SNR. Without a reference, the QPSK global-phase ambiguity gives $d_{\min}^{(E)} = 0$ and an SER near $0.75$ throughout the simulated range. All four equal-power reference-assisted designs remove this ambiguity and exhibit similar SER waterfalls. With $40000$ user-symbol decisions per point, their SERs at $-3$ dB are $4.00 \times 10^{-4}$, $2.25 \times 10^{-4}$, and $2.75 \times 10^{-4}$ for the constant, random, and uniform patterns, respectively. No errors are observed for these designs from $-1$ to 3 dB; the corresponding points are displayed at the half-event level $1.25 \times 10^{-5}$. For interpretation, the normalized minimum energy distances are $8.95$, $9.38$, and $9.79$, respectively. The uniform pattern attains the largest distance and is also selected by the finite codebook search, so the uniform and optimized curves coincide.

To test whether this distance predicts more than the five representative designs, Fig. 8 uses a 32-entry equal-power reference codebook with different phase and spatial-support patterns. At $-6$ dB, the correlation between the normalized minimum distance and the logarithm of the ML SER is $-0.98$. Thus, increasing the worst-case intensity separation systematically lowers the detection error, directly supporting the max–min criterion in (84).

Finally, Fig. 9 uses the same optimized reference to compare QPSK, 16-QAM, 64-QAM, and 256-QAM. As the modulation order increases, the denser constellation reduces the separation between intensity codewords and shifts the SER curve toward higher SNR. Over the sampled SNR grid, the SER first falls below $10^{-2}$ at $-3$, 3, 9, and 15 dB for QPSK, 16-QAM,

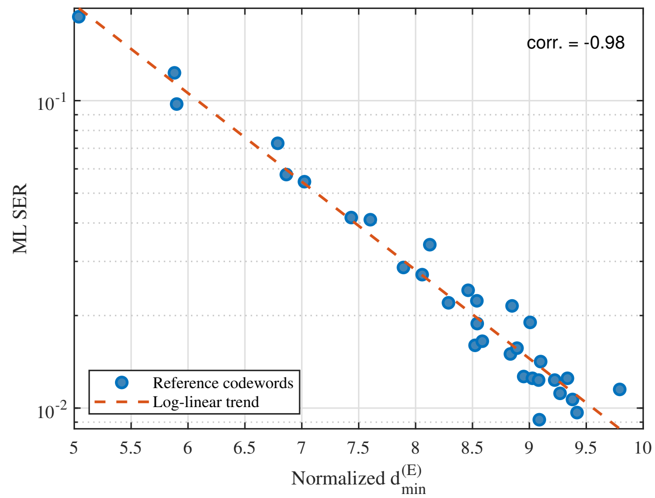


Fig. 8: ML SER versus normalized energy-domain minimum distance over an equal-power reference codebook.

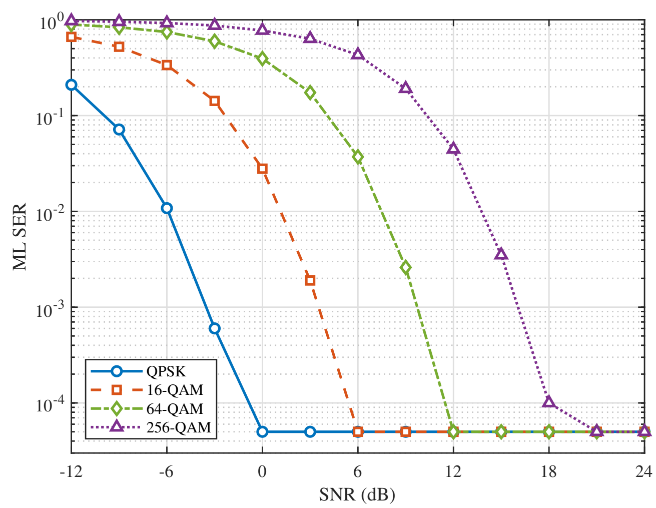


Fig. 9: ML SER for QPSK, 16-QAM, 64-QAM, and 256-QAM under perfect CSI. Zero-error points are displayed at the half-event level.

64-QAM, and 256-QAM, respectively. Thus, the reference removes the structural phase ambiguity for all four constellations, whereas the remaining SNR requirement is governed by finite-alphabet energy separation.

### C. End-to-End Symbol Detection with Estimated CSI

We now construct $\mathbf{G}_{\mathrm{d}}$ from the channel estimate obtained in the training phase, while retaining the two-user geometry and four-state data reference used above. Fig. 10 compares the three detectors when the training and data SNRs are equal. At $-6$ dB, the channel NMSE is $0.56$ dB and the estimated-CSI SERs of ML, projected GN, and reference-assisted GS are $1.98 \times 10^{-1}$, $1.88 \times 10^{-1}$, and $2.13 \times 10^{-1}$, respectively, compared with $1.04 \times 10^{-2}$, $1.46 \times 10^{-2}$, and $2.08 \times 10^{-2}$ under perfect CSI. At $-3$ dB, the channel NMSE decreases to $-1.76$ dB and the corresponding gap narrows to approximately one order of magnitude. Thus, channel mismatch, rather than

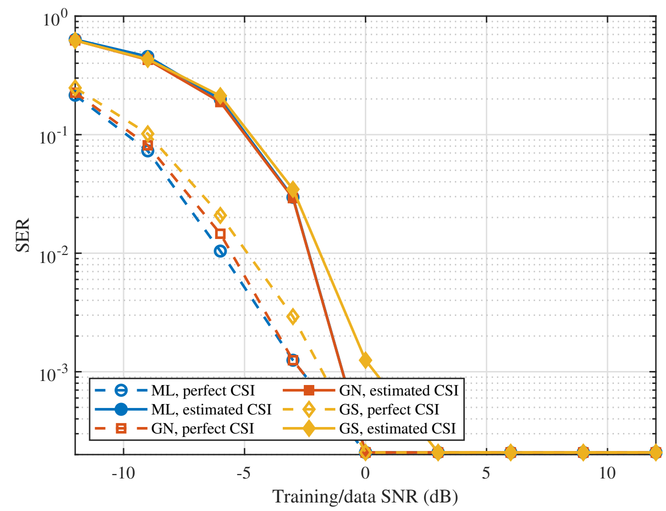


Fig. 10: SER under perfect and estimated CSI for different symbol detectors.

the choice among the three detectors, is the main limitation in the low-SNR region.

The two stages are separated in Fig. 11. In Fig. 11(a), the data SNR is fixed at $-6$ dB while the training SNR is varied. Without a training reference, the SER remains near $0.75$ because the channel phase is unidentifiable. The four reference-assisted designs instead reduce the SER from approximately $0.20$ at $-6$ dB to approximately $10^{-2}$ at 18 dB, where data-phase noise becomes dominant. In Fig. 11(b), a common channel ensemble estimated at a training SNR of 12 dB is used while the data SNR is varied. The no-reference SER again remains near $0.75$, whereas the reference-assisted SERs decrease from $0.21$–$0.23$ at $-12$ dB to $1.05 \times 10^{-2}$–$1.47 \times 10^{-2}$ at $-6$ dB. The optimized training reference is distinct, while the uniform and optimized data references coincide and are therefore represented by a single curve in panel (b). These results separate the roles of the two references: the training reference enables reliable CSI recovery, while the data reference ensures finite-alphabet separability.

### D. Effect of Reference and RIS Measurement Diversity

We finally examine how spatial and reference-state diversity change the finite-alphabet decision margin. To compare configurations with different numbers of observations, $d_{\min}^{(E)}$ is normalized by the root-mean-square noiseless intensity. The two-user QPSK setting and four-phase reference are used unless otherwise stated.

Fig. 12 varies the square RIS from $2 \times 2$ to $8 \times 8$. At $-12$ dB, increasing $N_{\mathrm{r}}$ from 4 to 64 raises the normalized minimum distance from 1.68 to 9.61 and reduces the ML SER from 0.641 to 0.220. The nearly monotonic correspondence between the two panels confirms that the element outputs act as distinct spatial measurements rather than as a single accumulated-energy statistic.

Reference-state diversity produces the analogous trend in Fig. 13. The reference phases follow the four-phase cycle in (79); hence every added state provides another projection

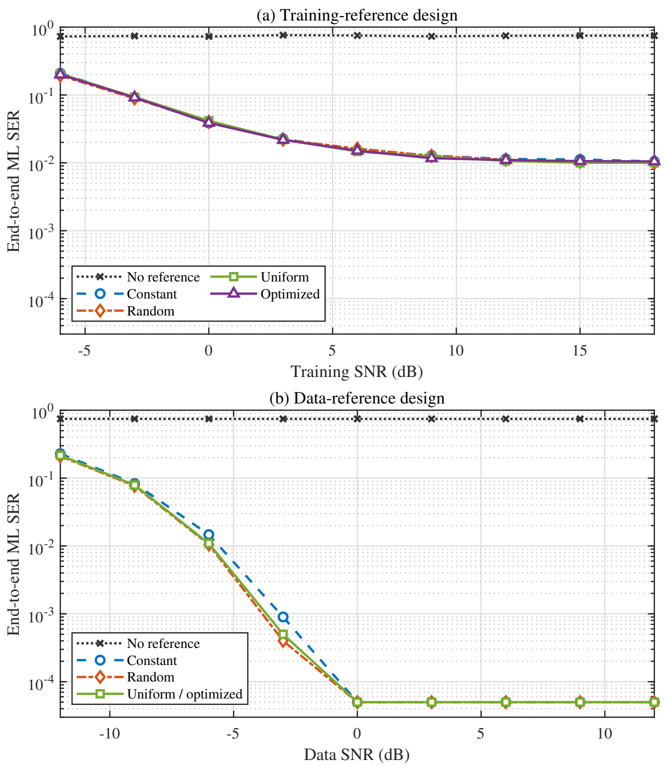


Fig. 11: End-to-end ML SER versus (a) training SNR for different training-reference designs at a data SNR of $-6$ dB and (b) data SNR for different data-reference designs at a training SNR of 12 dB. Zero-error points are displayed at the half-event level.

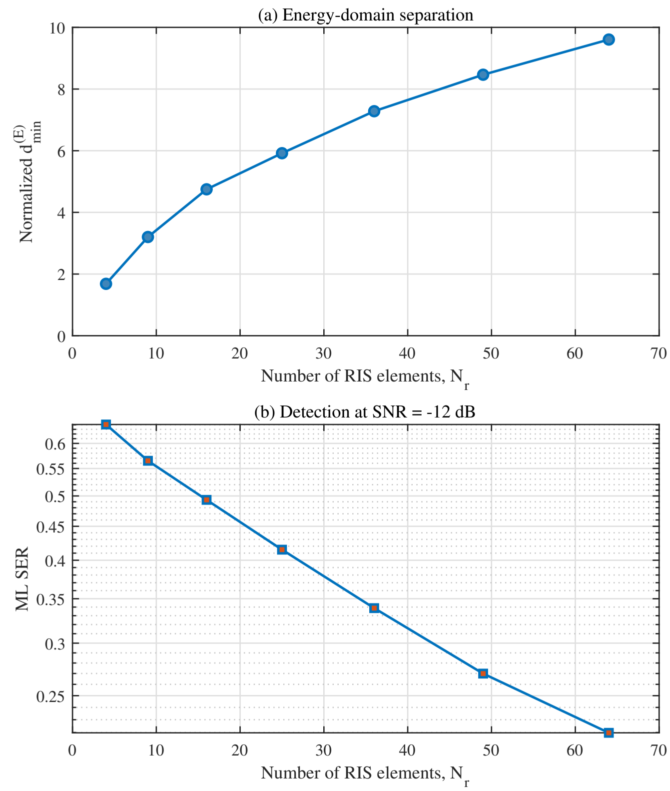


Fig. 12: Effect of the number of RIS element-level observations on (a) the normalized energy-domain minimum distance and (b) ML detection.

but also adds $N_\mathrm{r}$ intensity samples. As $R$ increases from 1 to 8, the normalized minimum distance grows from 2.75 to 13.58, while the SER at $-12$ dB falls from 0.495 to 0.0798. The gain is therefore accompanied by a linear increase in measurement and reference-switching overhead.

Fig. 14 connects spatial conditioning to end-to-end detection. With a $0.25^\circ$ user separation, the steering-vector correlation is 0.9995 and $\sigma_{\min}(\mathbf{V}/\sqrt{N_\mathrm{r}}) = 0.0214$; the perfect- and estimated-CSI SERs are consequently 0.194 and 0.259, respectively. At $8^\circ$, these manifold metrics become 0.606 and 0.628, and the SERs decrease to $1.77 \times 10^{-2}$ and $2.57 \times 10^{-2}$. Near $16^\circ$, the normalized minimum singular value reaches 0.990 and the estimated-CSI SER is $1.23 \times 10^{-2}$. The small rebound beyond this point is the finite-aperture sidelobe behavior rather than a failure of the conditioning argument. Here the training SNR is 12 dB, the data SNR is $-6$ dB, and $T_\mathrm{p} = 16$.

Finally, Fig. 15 compares the two forms of diversity at equal overhead. Both collect 64 samples through four reference states and 16 element outputs per state. The reference-only design repeats one fixed $4 \times 4$ subarray, whereas the joint design selects four state-dependent element masks by maximizing $d^{(E)}_{\min}$ over a 4096-entry equal-power codebook. Joint design increases the normalized minimum distance from 4.75 to 4.84 and reduces the SER from 0.137 to 0.126 at $-4$ dB and from 0.0580 to 0.0510 at $-2$ dB. The improvement is modest but consistent, showing that changing $\mathbf{G}$ is useful only when the measurement states are selected to complement the reference projections.

These results support the two-part design principle in Section III: the RIS measurement matrix must first preserve relevant field differences, after which reference-phase diversity projects those differences into separated intensity observations.

## VI. Conclusion

This paper presented a comprehensive framework for signal recovery in holographic RIS receivers using only amplitude measurements. By employing a known reference wave and formulating the recovery as a nonlinear least-squares problem, we enable the estimation of both amplitude and phase of the transmitted signal. The proposed Gauss-Newton algorithm efficiently solves this optimization, with simulations demonstrating accurate recovery under various SNR conditions.

The mathematical derivation provides complete details from signal transmission through the channel, interference with the reference wave at the RIS, intensity measurement, to the final recovery algorithm. This framework serves as a foundation for more advanced RIS receiver designs, including those with adaptive reflection coefficients, multiple users, and frequency-selective channels.

Future work will extend this approach to practical implementations with hardware impairments, investigate deep learning-based recovery for reduced complexity, and explore applications in joint communication and sensing with RIS.

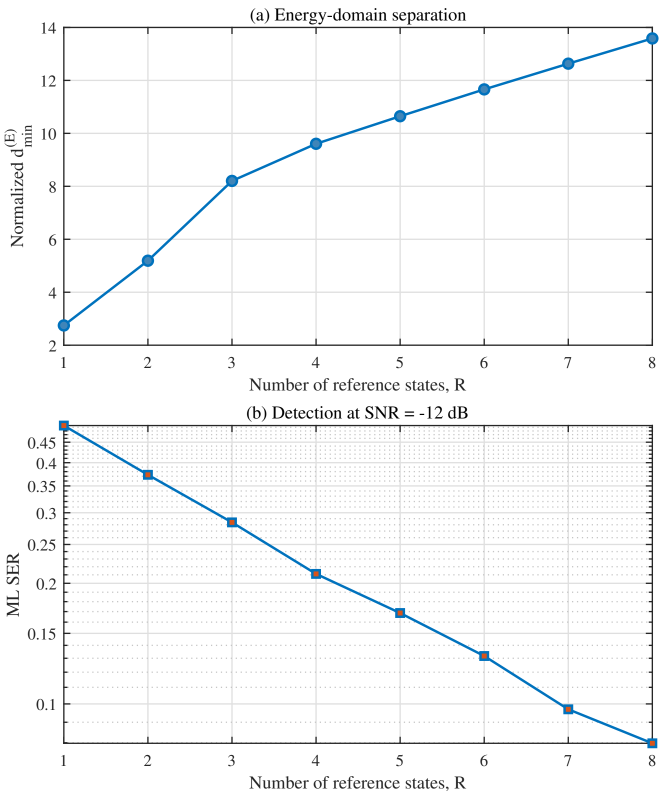


Fig. 13: Effect of the number of phase-shifted reference states on (a) the normalized energy-domain minimum distance and (b) ML detection.

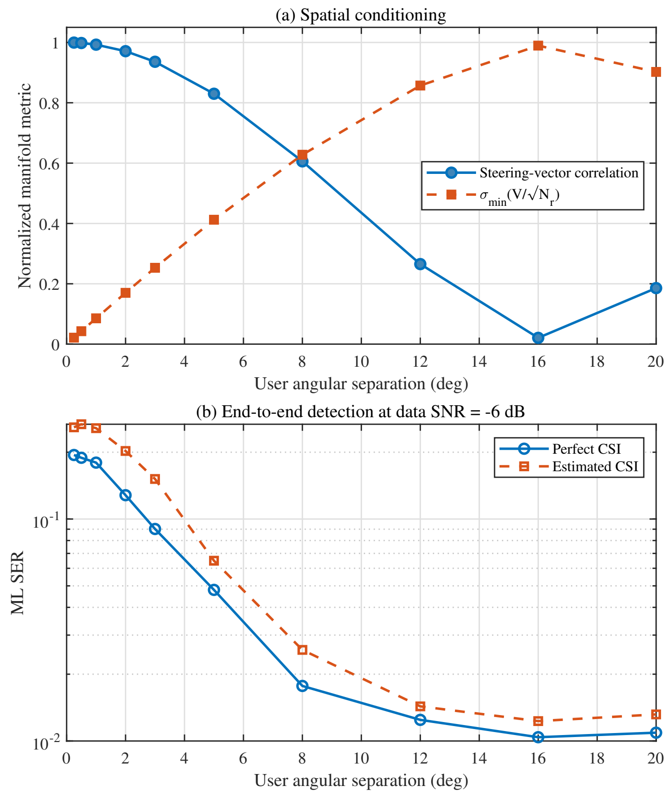


Fig. 14: Effect of user angular separation on (a) array-manifold conditioning and (b) end-to-end ML detection.

## Appendix A

## Appendix B

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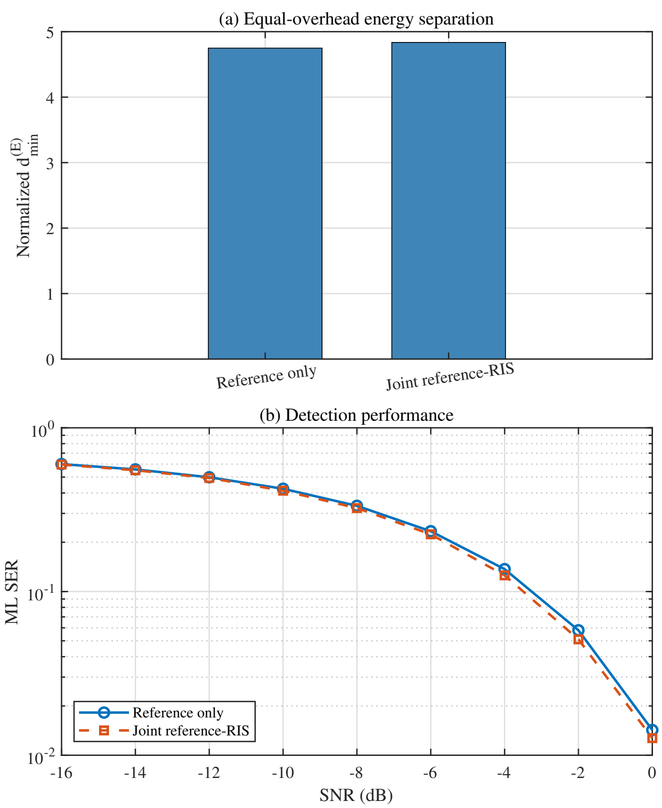


Fig. 15: Equal-overhead comparison of reference-only and joint reference-RIS measurement diversity: (a) normalized minimum distance and (b) ML SER.